EModel-Seek: Sequence Identification for Structure-First Cryo-EM Structural Discovery via Contrastive Learning

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Abstract

Cryo-electron microscopy (cryo-EM) enables a structure-first approach to biological discovery in which macromolecular assemblies are visualized before their molecular identities are established. This approach requires sequence information to be recovered from density maps without a known target sequence or a supplied backbone model. Here, we present EModel-Seek (EModelS), a method that extends EModelX with supervised contrastive learning for amino acid type prediction and downstream sequence database searches. EModelS can be combined with state-of-the-art structure-first protocols to identify unknown proteins directly from density maps when sequences are unavailable. On 99 experimental cryo-EM maps at 2–4 $\mathring{A}$ resolution, EModelS achieves a mean residue-type accuracy of 0.546, compared with 0.484 for EModelX. Supervised contrastive learning yields more separable residue-type embeddings, consistent with this accuracy gain. Across 206 evaluated chains, EModelS achieves a mean reference-normalized sequence identity of 0.836, compared with 0.783 for sequence-free EModelX. Under a joint criterion of sequence identity and query coverage both ≥ 0.90, EModelS recovers qualifying chains in 44% of complexes, versus 39% for sequence-free EModelX, 38% for ModelAngelo and 32% for findMySequence. Case studies further show that continuous chain traces improve database matching relative to fragmented models, and that density-supported residue alternatives can guide local model refinement after sequence search. Together, these results support EModelS as a method for linking cryo-EM map interpretation to sequence-based protein identification.

1 Introduction

Cryo-electron microscopy (cryo-EM) has transformed structural biology by enabling the visualization of macromolecular assemblies at near-atomic resolution without the need for crystallization [1, 2, 3, 4]. Accurate atomic model building from cryo-EM density maps is fundamental to elucidating molecular mechanisms, facilitating structure-guided drug discovery [5], and advancing structural genomics [6, 7]. Sequencing technologies now provide protein sequences at unprecedented scale [8, 9]. Structural studies have largely continued to follow a sequence-driven, top-down paradigm in which a biological target is typically identified and its sequence established before its structure is determined, a practice shaped by long-standing experimental workflows as well as by ready access to sequence information [10, 11].

Recent developments such as CryoSeek [10] have introduced a practical workflow for an emerging structure-first paradigm of structure-guided biological discovery: first observing macromolecular assemblies and then determining their molecular identities. By enabling direct visualization of proteins and complexes in heterogeneous or native biological samples, cryo-EM offers opportunities to uncover previously uncharacterized proteins without requiring their identities to be known in advance. CryoSeek combines automated model building from cryo-EM density maps with Foldseek [12]–based structure matching, linking experimental densities to candidate proteins through searches against structural databases such as the AlphaFold Protein Structure Database (AFDB) [13]. Although structure-based comparison methods [14, 15] have proven powerful for homology detection and functional inference, their effectiveness depends on database coverage. Moreover, proteins with substantially different sequences can share similar folds [14, 16, 17], making structural similarity alone insufficient to establish the exact sequence of an observed protein. Candidate matches therefore often require residue-by-residue inspection and manual correction by experienced structural biologists [18, 19]. Such time-consuming, expertise-intensive curation remains a major bottleneck in large-scale cryo-EM map interpretation.

An alternative and potentially more general approach is to infer amino acid sequence information directly from cryo-EM density maps, thereby enabling sequence-based protein identification against comprehensive repositories such as UniProt [20]. This strategy could provide a direct link between map interpretation and molecular identity. Existing methods such as `findMySequence` [21] and EMSequenceFinder [22] estimate amino acid type probabilities from pre-built backbone models and then perform probabilistic matching against sequence databases. ModelAngelo [23] further couples automated atomic model building with residue-type probabilities for HMM-based protein identification from cryo-EM maps. However, the requirement for accurate backbone traces can be difficult to meet for unknown complexes, where chain segmentation and atomic connectivity remain uncertain.

Our previous work, EModelX [24], enables automated atomic model building from cryo-EM density maps given the corresponding protein sequences. Its deep learning approach predicts atom positions and amino acid identities from volumetric density data to support model building. Building on EModelX, we have also developed methods for CryoSeek-related applications, including automated glycoprotein modeling [25] and handedness determination [26]. However, the identification of unknown proteins within the CryoSeek workflow still calls for a robust, automated method that recovers sequence information from density maps and uses it to establish protein identity. Accurate amino acid type assignment is central to this task, as errors in residue assignments can compromise both sequence recovery and subsequent database matching.

Here, we present EModelS, an enhanced deep learning method that extends our previous EModelX pipeline by incorporating supervised contrastive learning [27] to improve amino acid type prediction and sequence recovery. Supervised contrastive learning promotes inter-class separability and intra-class consistency by explicitly pulling together features of the same amino acid type while pushing apart those of different types [28], thereby refining the latent embedding space for amino acid types. EModelS further incorporates a protein identification module based on probabilistic sequence search, forming a unified pipeline for *de novo* sequence identification directly from cryo-EM density maps. By coupling voxel-level discriminative representation learning with downstream sequence searches, EModelS links cryo-EM map interpretation to protein identification, providing an automated approach for discovering unknown proteins from experimental data.

2 Results

2.1 EModelS combines voxel-wise classification with supervised contrastive learning

EModelS assigns an amino acid type to each voxel of a cryo-EM density map by training residue classification together with a supervised contrastive objective (Figure 1). The map is padded, cut into patches and preprocessed before it enters a residual 3D U-Net. The decoder features are used in two ways. A final convolution and softmax predict residue types, with supervision provided by cross-entropy against structure-derived labels. The same features are sampled at atom positions, normalized, and optimized to cluster features of the same amino acid type and separate those of different types (Supplementary Fig. S1). The two losses are added with equal weight. Each training label is made by spreading an atom's residue type into nearby voxels: the contribution is strongest at the atom, weakens with distance, and stops at the map resolution. In overlapping regions, the strongest contribution sets the type; a voxel that contains an atom keeps that atom's type. These labels are used only in training. Inference reads the density map alone.

2.2 EModelS improves residue-type prediction and feature representation

We compared EModelS with EModelX on 99 cryo-EM maps using the deposited-residue denominator defined in Methods. EModelS has a higher per-map residue-type accuracy than EModelX in panel a, with mean values of 0.546 and 0.484, respectively (Figure 2a). Across the four resolution intervals, EModelS remains above EModelX, showing that the improvement is not restricted to a single resolution range (Figure 2b). The paired map-level comparison in panel c is concentrated above the diagonal: EModelS improves the residue-type accuracy in nearly every case, with only a small number of cases showing little or no gain.

The feature analyses visualize the representation-level contribution of supervised contrastive learning. For same-type and different-type residue pairs, EModelS increases the median separation between the two cosine-similarity distributions relative to EModelX, with same-type similarities rising more than different-type similarities (Figure 2d). The corresponding t-SNE projections show clearer organization by amino acid type in the EModelS feature space, consistent with the higher nearest-neighbour accuracy annotated in the panel (Figure 2e). Together, these analyses corroborate the accuracy gains by showing improved residue-feature separability at the

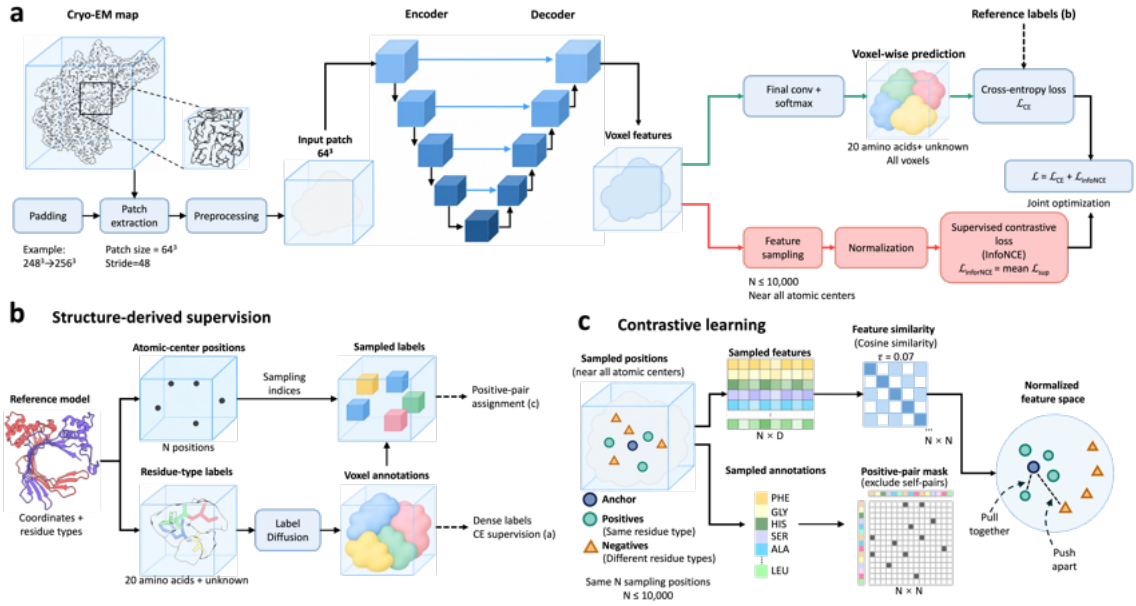


Figure 1: **Training workflow for residue-type prediction in EModelS. a,** Overall network framework. Density-map preparation includes padding, patch extraction and preprocessing; a residual 3D U-Net produces decoder features that feed two branches: a final convolution and softmax for voxel-wise residue classification, trained by cross-entropy ($\mathcal{L}_{\mathrm{CE}}$), and a branch that samples features at atom positions, normalizes them and applies an InfoNCE-based supervised contrastive loss ($\mathcal{L}_{\mathrm{sup}}$). The combined objective is $\mathcal{L} = \mathcal{L}_{\mathrm{CE}} + \overline{\mathcal{L}_{\mathrm{sup}}}$. **b,** Training labels. Each atom spreads its residue type into nearby voxels ("Label Diffusion"), strongest at the atom and fading out at the map resolution. Where these regions overlap, the stronger residue is kept, and a voxel that contains an atom keeps that atom's type. **c,** Supervised contrastive loss. Atom-associated decoder features are randomly subsampled to at most 10,000 sites, filtered by a residue-type mask and assembled into a feature matrix; after L2 normalization, each retained site serves as an anchor whose InfoNCE loss is computed from the pairwise similarity matrix, pulling same-type features together and pushing different-type features apart.

representation level.

The per-type changes are heterogeneous (Figure 2f). Methionine, aspartate and cysteine show the largest gains for EModelS and were among the lowest-recall residue types under EModelX (Supplementary Fig. S2). Valine is the only residue type with lower recall in EModelS. This pattern is consistent with the structural similarity of valine to other hydrophobic residues in cryo-EM density, whereby improvements in the global embedding margin can redistribute errors among closely confusable classes—an effect that is not resolved by the aggregate per-type summary alone. Overall, the rise in amino-acid accuracy aids sequence search (Supplementary Figs. S3 and S4; Supplementary Table S1), which we evaluate next across methods and structural complexity.

2.3 Sequence identification across methods and structural complexity

We evaluated sequence identification on a common set of 206 chains from 99 complexes (Figure 3). EModelX denotes the sequence-free configuration, EModelX (w/o seq), and ModelAngelo was run with version 1.0.12. The mean chain-level reference-normalized sequence identity is 0.836 for EModelS, 0.783 for EModelX (w/o seq), 0.816 for ModelAngelo and 0.561 for `findMySequence` (Figure 3a). The EModelS and ModelAngelo distributions have similar medians, both close to complete identity, while EModelS has the highest mean across the evaluated chains. Chain-wise `phmmer` comparisons of EModelX(s) and EModelS(s), stratification by deposited chain length, and the full 206-chain listing are given in Supplementary Figs. S5 and S6 and Supplementary Table S2.

Panel b compares the per-complex mean sequence identity of EModelS with ModelAngelo. The number of complexes favouring EModelS is substantially larger than the number favouring ModelAngelo, indicating that the advantage is distributed across many complexes rather than driven by a single example. At the complex level, panel c uses the criterion that sequence identity and query coverage are both ≥ 0.90. It classifies a complex as *Fully matched* when all evaluated chains satisfy both thresholds and as *Partially matched* when only a subset do. EModelS has more fully and partially matched complexes than the comparison methods, showing that its advantage extends beyond the average chain-level sequence identity to multi-chain assemblies.

Panels d and e report hit rate, where a hit is a chain satisfying sequence identity and query coverage both ≥ 0.90. Hit rate decreases as map resolution worsens for all methods, and EModelS remains above ModelAngelo and `findMySequence` in the poorer-resolution bins. Chain length has a dual effect: longer queries supply more sequence evidence for database matching, yet continuous tracing of long chains is harder and automatic builders more readily split them into many short fragments, so each search query covers only part of the deposited sequence (Supplementary Fig. S7). A concrete example of how such fragmentation weakens identification is given in the 7DEG case study below (Figure 4). EModelS leads the comparison methods for chains longer than 400 residues. In the structure-first paradigm exemplified by CryoSeek, native samples more readily yield moderately resolved maps of large protein assemblies than small, high-resolution reconstructions, making identification on such targets particularly consequential. Panel f examines the Hard subset of 30 chains that combine resolution ≥ 3.5 Å with length > 400 residues, where poor density and chain fragmentation can compound; the complementary Other subset contains the remaining 176 chains. On Hard chains, EModelS attains a hit rate of 0.43, compared with 0.30 for ModelAngelo

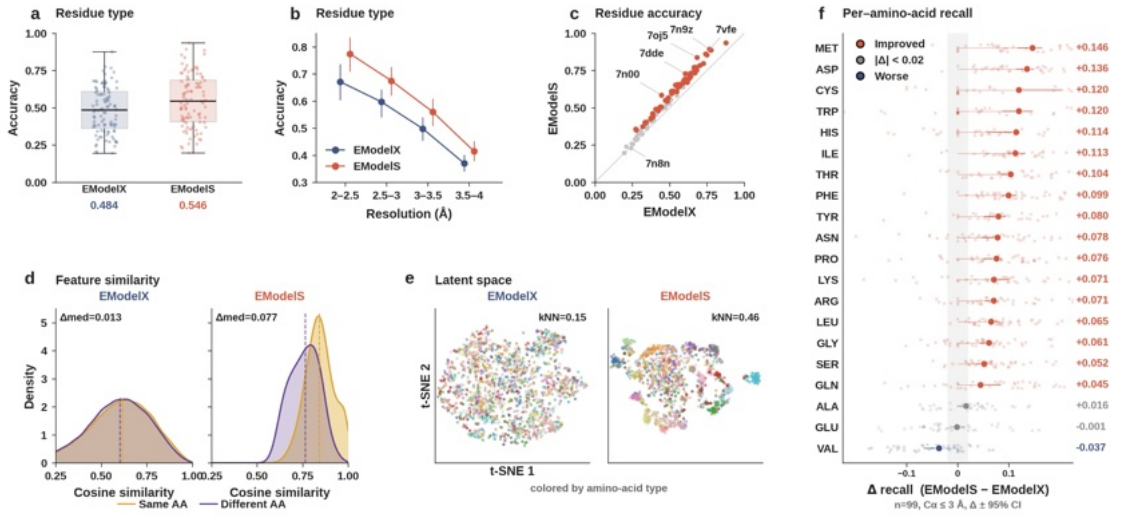


Figure 2: **Residue-type prediction and feature representation. a,** Per-map residue-type accuracy for EModelX and EModelS. Boxes show the interquartile range, the centre line the median, and whiskers 1.5× the interquartile range; outliers are omitted. Jittered points are individual maps, and the numbers beneath the boxes are mean accuracies (0.484 and 0.546). **b,** Mean accuracy across the four map-resolution intervals spanning 2–4 Å; EModelS is above EModelX in each interval. **c,** Paired per-map accuracies, with EModelX on the horizontal axis and EModelS on the vertical axis; the diagonal denotes equal performance and the points are predominantly above it. **d,** Cosine-similarity distributions for same-type and different-type residue features. EModelS increases the median same-versus-different separation relative to EModelX, with same-type similarities rising more than different-type similarities. **e,** t-SNE projections of residue features coloured by amino acid type. The projections visualize the clearer organization by amino acid type learned by EModelS; the annotated k-nearest-neighbour accuracies provide the corresponding quantitative summary. **f,** Per-type recall difference Δ = EModelS − EModelX. Large solid points are pooled residue-level Δ values (correct counts summed over maps, divided by the corresponding deposited-residue denominators). Horizontal bars are 95% bootstrap confidence intervals of the unweighted per-map mean Δ (4,000 resamples). Faint points are per-map Δ values (at most 40 shown per type). Methionine, aspartate and cysteine show the largest gains, whereas valine shows a decrease.

and 0.20 for `findMySequence`, a larger margin than on the Other subset (0.36, 0.33 and 0.23, respectively).

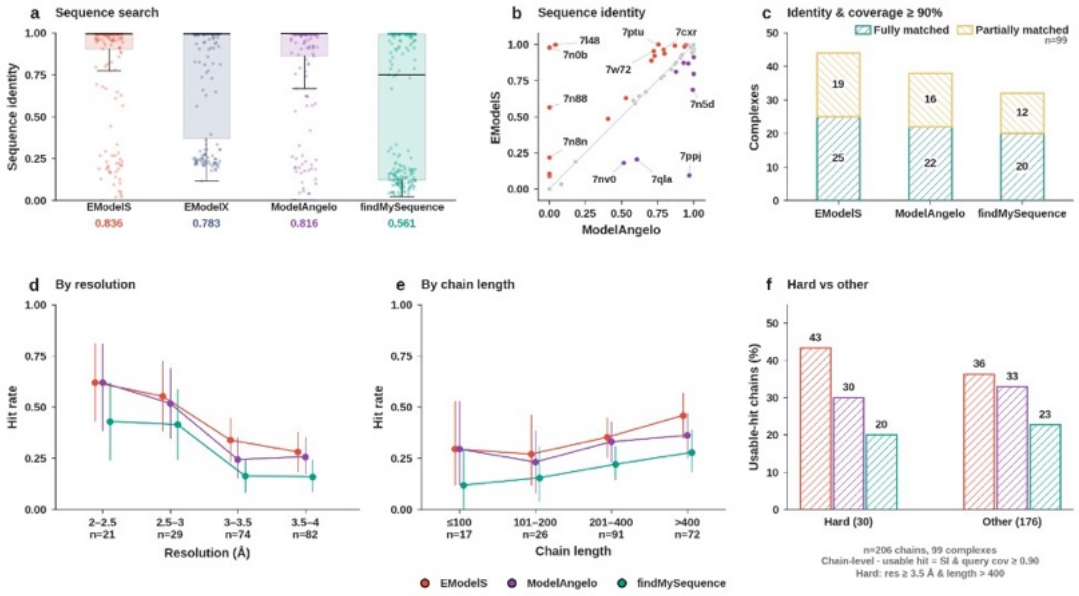


Figure 3: **Sequence-identification benchmark on 206 chains from 99 complexes. a,** Distributions of reference-normalized sequence identity for EModelS, EModelX (w/o seq), ModelAngelo and `findMySequence`; the EModelS and ModelAngelo medians are both close to complete identity. **b,** Per-complex mean sequence identity for EModelS versus ModelAngelo. The diagonal indicates equal performance, and the number of points favouring EModelS is substantially larger. **c,** Complex-level outcomes using sequence identity and query coverage both ≥ 0.90. "Fully matched" means that all evaluated chains satisfy both thresholds; "Partially matched" means that only a subset do. Complexes with no qualifying chain are omitted from this panel. **d,e,** Hit rate, defined as the fraction of chains satisfying sequence identity and query coverage both ≥ 0.90, stratified by map resolution and chain length. **f,** Chain-level hit rate for Hard chains (resolution ≥ 3.5 Å and length > 400 residues; $n = 30$) and the complementary Other subset ($n = 176$). EModelS, ModelAngelo and `findMySequence` are shown consistently in red, purple and teal; EModelX (w/o seq) is additionally included in panel a.

2.4 Chain completeness governs sequence identification in a 7DEG example

The 7DEG case study examines how differences in automatic model building translate into sequence identification for the same target chain (Figure 4). Panel a compares the deposited reference with the models built by ModelAngelo and by EModelS. Both automatic models recover a fold that closely matches the ground-truth structure in the displayed region, indicating that the subsequent difference in sequence search is not due to a major failure of either method to interpret the density.

The decisive distinction lies in chain completeness. ModelAngelo builds the displayed chain as two separate fragments (model chains g / s), whereas EModelS returns a more continuous chain for the same region (model chain H). Because sequence searching uses the built chain as the query, fragmentation shortens the contiguous sequence evidence available for database matching and can weaken or prevent a high-identity hit even when the local backbone geometry is essentially correct. Panels b and c show this consequence directly in sequence alignments against the deposited chain E / B class (147 residues): from the fragmented queries, ModelAngelo returns UniProt `O67936` as its top hit, with reference-normalized sequence identity 0.578 and query coverage 1.000 (aligned length 85; the hit covers only the C-terminal portion of the deposited chain), whereas from the continuous query (chain H, reverse), EModelS returns `G5DGC8` as its top hit, with sequence identity 1.000, reference coverage 1.000 and query coverage 0.987. A continuous query better preserves contiguous residue evidence; fragmented queries yield shorter alignments that are harder to score as confident identifications. Across the 99-map benchmark, ModelAngelo likewise tends to build shorter chains than EModelS. The same fragmentation tendency also helps explain why ModelAngelo falls further behind EModelS on longer chains, where unbroken connectivity matters more for assembling a searchable query. Beyond improving amino-acid type accuracy, this case suggests that recovering high-confidence chain connectivity is also important for sequence identification, even though doing so remains difficult when structures are predicted directly from density.

2.5 EModelS supports sequence-guided refinement in CryoSeek case studies

The three examples in Figure 5 originate from sequence-free EModelS models generated within the CryoSeek workflow. They test whether EModelS can turn density-derived residue predictions into sequence information that supports subsequent model building. All three cases were searched against the same local copy of UniProtKB/TrEMBL (UniProt release 2025_03, 18 June 2025). Across these cases, EModelS recovers a near-identical database sequence for one map (9XFV), whereas the other two return closest available homologs rather than exact matches. Supplementary Table S3 and Supplementary Figs. S8 and S9 summarize the shared top UniProt hits, sequence-free rebuild metrics and per-chain identities; the corresponding three-track alignments of the EModelS network sequence, the searched UniProt sequence and the deposited sequence are shown in Supplementary Figs. S10–S12.

For 9XFV, among retained UniProt hits, the best match to the deposited chain reaches 98.3% reference-normalized sequence identity (identical aligned residues divided by deposited chain length), providing a direct sequence identification from the density-derived model. Panel b shows the sequence-free EModelS Cα model (chain W) superimposed on the deposited 9XFV protomer (chain R) at this pre-search stage; returning a retrieved sequence to guide subsequent model building is considered for the homolog cases below. For this single-chain Cα superposition, TMalign gives an RMSD of approximately 0.73 Å and a TM-score of approximately 0.964, normalized by the deposited chain length. Thus, the sequence-free Cα model closely reproduces the protomer fold while already providing sequence information sufficient to retrieve a near-identical database hit.

For 9XNC and 9XNH, the closest retrieved homologs have reference-normalized sequence identities of 0.602 (`A0A4Q7Z3W7`) and 0.457 (`A0A0D7KGF4`), respectively. When the local database contains no sequence sufficiently close to the deposited sequence, the retrieved homolog is therefore not identical to the deposited sequence, although it can still be the closest available candidate. This

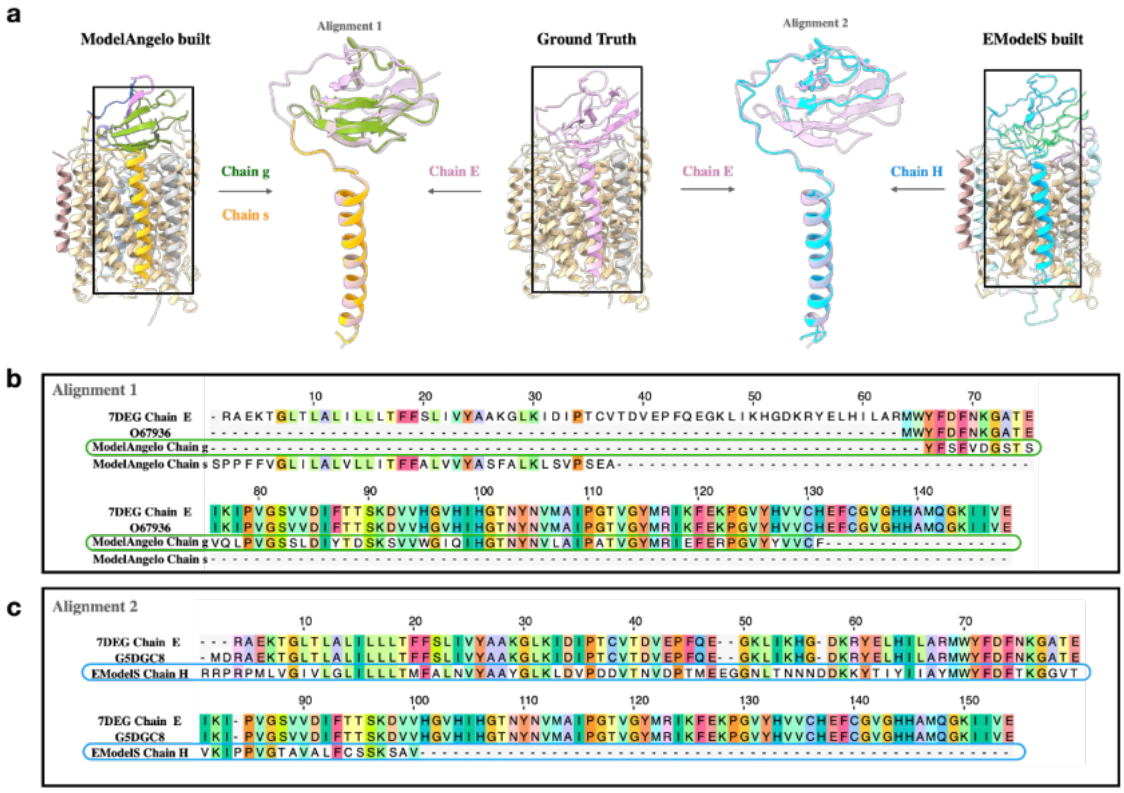


Figure 4: **Automatic model building and sequence alignment for 7DEG. a,** Structural comparison of the deposited reference (chain E) with the ModelAngelo and EModelS models for the same region. Both automatic models closely match the ground-truth fold; ModelAngelo builds the chain as separate fragments (chains g / s), whereas EModelS yields a more continuous chain (chain H). **b,** Sequence alignment of the ModelAngelo top hit (`O67936`) with the deposited chain; sequence identity 0.578, query coverage 1.000. **c,** Sequence alignment of the EModelS top hit (`G5DGC8`) with the deposited chain; sequence identity 1.000, query coverage 0.987. Beyond amino-acid type accuracy, the case highlights the importance of recovering high-confidence chain connectivity for sequence identification.

situation is expected in structure-first CryoSeek discovery, where the observed protein may be absent from existing sequence databases. Limited database coverage need not preclude further model refinement: even a confidently retrieved homolog can disagree with the map at individual sites, because the observed protein may itself be a newly discovered sequence. Local mismatches can therefore be flagged and corrected by asking which amino-acid assignment better agrees with the density, reducing the impact of an imperfect database hit on the final model.

Panel c summarizes all bidirectional substitutions observed in 9XNC and 9XNH. Here, $X \rightarrow Y$ means that residue X in the searched sequence is replaced by residue Y in the deposited sequence; $Y \rightarrow X$ denotes the converse substitution at other aligned sites. Only amino-acid pairs observed in both directions and supported by at least four sites in total are shown. Restricting the comparison to bidirectional pairs controls for residue-type–intrinsic effects on the local map–model correlation (Phenix `CC_mask`) [29], because any systematic bias associated with a particular amino acid would reverse sign between the two directions. For each site, the plotted change is the deposited-model per-residue CC minus the searched-model per-residue CC. The deposited residue generally has the higher CC in both directions, indicating that the preference reflects agreement with the density rather than the chemical identity of the residue itself, and that density-supported residue alternatives can guide subsequent sequence and model refinement.

Panels d and e illustrate this refinement step with local examples. In panel d (9XNC), the searched residue at the highlighted site carries a side chain that is too large for the surrounding density; replacing it with the deposited amino acid raises the local CC. In panel e (9XNH), the searched residue is instead too small, leaving unused density that is better filled by the deposited amino acid, again with an increase in CC. Taken together, these examples show that mismatches between a retrieved homolog and the true sequence can be flagged and corrected by comparing local map–model correlation after candidate substitutions, providing a computational route to the final density-supported optimization of residue assignments. A site-resolved analysis of all 143 deposited-versus-searched mismatch positions, including per-residue CC comparisons and rank among the 20 amino acids, is provided in Supplementary Fig. S13. This observation aligns with the CryoSeek strategy of first identifying a plausible homolog and then refining the model around density-supported residue assignments.

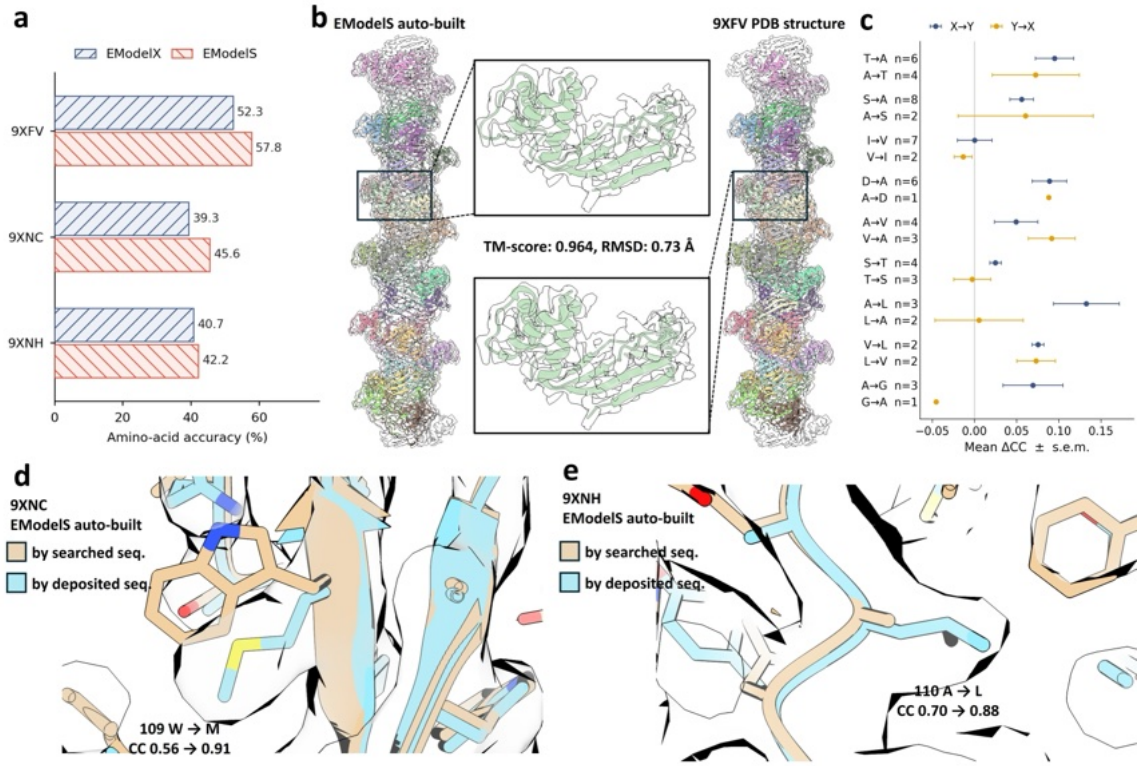


Figure 5: **Sequence-guided structural assessment in three CryoSeek case studies. a,** Amino-acid accuracy for EModelX (w/o seq) and EModelS in the cases labelled 9XFV, 9XNC and 9XNH. **b,** The sequence-free EModelS Cα model (chain W) and the deposited 9XFV protomer (chain R). The single-chain Cα superposition gives a TMalign RMSD of approximately 0.73 Å and a TM-score of approximately 0.964 normalized by deposited chain length. **c,** Mean CC changes for bidirectional searched-to-deposited amino-acid substitutions in 9XNC and 9XNH. Bars show the deposited-model per-residue CC minus the searched-model per-residue CC, with standard errors. **d,** Local example in 9XNC where the searched side chain is oversized relative to the density; substitution to the deposited residue increases the local CC. **e,** Local example in 9XNH where the searched side chain is undersized; substitution to the deposited residue likewise increases the local CC.

3 Discussion

EModelS improves residue-type prediction and sequence identification from cryo-EM density maps without requiring a target sequence as input. The residue-level analyses show higher accuracy across resolution intervals, consistent per-map gains, and more separated feature representations after supervised contrastive learning. The sequence benchmark places these gains in a comparative setting against EModelX (w/o seq), ModelAngelo and `findMySequence`: EModelS attains higher chain-level sequence identity and recovers more fully and partially matched complexes. The advantage persists for long chains in lower-resolution maps, where continuous tracing and sequence evidence are both harder to obtain.

The CryoSeek case studies illustrate how this capability can support a structure-first discovery workflow. A sequence recovered from a density-derived model can be supplied to sequence-guided EModelS for subsequent model building, and the closest retrieved homolog provides a practical starting point for candidate assessment when an exact match is unavailable. Residue-type gains also help when a target sequence is already available: higher amino-acid accuracy improves local chain assignment and continuity during sequence-guided building (Supplementary Fig. S14). The bidirectional substitution analysis, together with the oversized and undersized side-chain examples in Figure 5d,e, further points to a density-informed refinement loop—updating residue assignments by local correlation, rebuilding the atomic model and reassessing sequence identity—that can be assisted computationally and aligns with the homolog-search-then-refine strategy used in CryoSeek.

Three considerations follow from density-derived sequence identification in a structure-first workflow. First, as with any sequence-search approach, identification is bounded by database coverage: when the observed protein is absent from the searchable repository, the returned hit is the closest available homolog rather than an exact identity. Second, complete complex-level identification remains a stricter requirement than single-chain retrieval, because every constituent chain must meet the joint sequence-identity and coverage criterion. Third, short or fragmented modeled chains can hinder identification even when residue types are predicted. The 7DEG case suggests that fragmentation shortens the contiguous query and can weaken the search even when the local fold is essentially correct, whereas a more continuous trace better preserves the sequence evidence needed for a high-identity hit. Relative to ModelAngelo, EModelS produces far fewer modeled chains shorter than 30 residues, a potential advantage for future structure-identification tasks in which searchable queries need adequate contiguous length. Looking ahead, sequence search need not rely only on discrete residue-type assignments: probabilistic models of amino-acid type could be paired with probabilistic models of chain connectivity, so that uncertain peptide bonds are represented explicitly during search and automatic builders can better decide where to join or split segments. Such connectivity-aware searching would complement the residue-level gains reported here and may further improve identification on long or poorly continuous chains.

More broadly, recovering molecular identity directly from cryo-EM density constitutes a distinct problem for deep learning in structural biology, complementary to conventional sequence-guided model building. By establishing a curated benchmark of experimental maps together with explicit evaluation criteria for residue-type prediction and sequence identification, this work provides a common basis for method development. We hope that these resources will encourage further advances that accelerate structure-first discovery of previously uncharacterized proteins.

4 Methods

4.1 Datasets

High-resolution single-particle cryo-EM density maps were obtained from the Electron Microscopy Data Bank (EMDB) [30], each associated with a fitted atomic model in the Protein Data Bank (PDB).

For network training and validation, 6,082 and 89 maps released after January 2018 with reported resolutions between 1 Å and 6 Å were selected after removing entries with high sequence similarity. For independent testing, 99 single-particle maps released after January 2021 with resolutions between 2 Å and 4 Å were used, consistent with the benchmark employed in EModelX.

For supervision, each voxel in the experimental density map was labeled with its ground-truth amino acid type (20 standard residues plus an unknown class) and an atom-type label derived from the corresponding PDB structure, distinguishing (i) voxels proximal to any heavy atom, (ii) side-chain heavy-atom centres, (iii) non-Cα backbone centres (N, C or O) and (iv) Cα centres. All density maps were preprocessed by removing low-density voxels, clipping extreme intensity values, and normalizing by the maximum voxel intensity. From each processed map, cubic subvolumes of $64 \times 64 \times 64$ voxels were extracted with a stride of 48, with density patches used as network inputs and aligned label volumes providing the training targets.

4.2 Contrastive learning for amino acid type prediction

Accurate amino acid type prediction is critical for reliable sequence identification from cryo-EM maps. EModelS extends the U-Net [31] backbone of EModelX by augmenting voxel-wise classification with a supervised contrastive objective [27], jointly optimizing dense classification and residue-type feature separation.

Losses and objective. A label-smoothed cross-entropy (CE) loss provides dense supervision over $C = 21$ classes (20 standard amino acids and one unknown class). For voxel i with one-hot label $y_i \in \{0,1\}^C$ and predicted probabilities $\hat{y}_i$, the smoothed target is

$$\tilde{y}_{i,c} = (1-\varepsilon)\, y_{i,c} + \frac{\varepsilon}{C}, \qquad \varepsilon = 0.1,$$

and

$$\mathcal{L}_{\mathrm{CE}} = -\frac{1}{N} \sum_{i=1}^{N} \sum_{c=1}^{C} \tilde{y}_{i,c} \, \log(\hat{y}_{i,c}),$$

where empty / non-residue voxels are excluded (ignore-index training).

A supervised contrastive loss is applied to voxels at atomic centres (side-chain heavy atoms, non-Cα backbone N/C/O, or Cα). Let S_0 be that set in a mini-batch and

$$S = \begin{cases} S_0, & |S_0| \le N_{\max}, \\ \mathrm{UniformSample}(S_0, N_{\max}), & |S_0| > N_{\max}, \end{cases} \qquad N_{\max} = 10{,}000.$$

We compared three choices of S_0—all voxels, Cα centres only, and all atomic centres—and retained the all-atom centre set (about 5% higher absolute accuracy than the alternatives): all-voxel sampling

admits many background sites, whereas Cα-only sampling omits side-chain centres [32, 33, 27]. For an anchor $i \in S$ with L2-normalized embedding z_i, let $P(i) = \{p \in S \setminus \{i\} : \mathrm{label}(p) = \mathrm{label}(i)\}$ and $A(i) = S \setminus \{i\}$. The per-anchor loss is

$$\mathcal{L}_{\mathrm{sup}}(i) = -\frac{1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp\big(\mathrm{sim}(z_i, z_p)/\tau\big)}{\sum_{a \in A(i)} \exp\big(\mathrm{sim}(z_i, z_a)/\tau\big)}, \quad \mathrm{sim}(u, v) = \langle u, v \rangle,$$

with temperature $\tau = 0.07$ [34, 27]. Averaging over $S' = \{i \in S : |P(i)| > 0\}$ gives $\overline{\mathcal{L}_{\mathrm{sup}}}$. The joint objective is $\mathcal{L} = \mathcal{L}_{\mathrm{CE}} + \overline{\mathcal{L}_{\mathrm{sup}}}$, with both gradients back-propagated through the shared U-Net. Network parameters were optimized with Adam [35]. Ablation protocols that defer $\mathcal{L}_{\mathrm{CE}}$ to a frozen-encoder classification head are reported in Supplementary Table S4.

4.3 Sequence identification from cryo-EM maps

The pipeline schematic (model-derived queries, `phmmer`/`hmmsearch`, hit retention and reference comparison) is given in Supplementary Fig. S15; that figure also motivates reference-normalized Smith–Waterman identity over global Needleman–Wunsch scores. Metric definitions are in the evaluation subsections below.

We processed the 99 test maps with the EModelS U-Net to obtain voxel-wise amino acid probabilities, then built atomic models with the EModelX downstream modules without prior sequence information. Chains with length $L \geq L_{\min}$, $L_{\min} = 30$, were retained.

Residue-type log-probabilities at Cα positions were converted into profile HMMs in both forward and reverse directions. With per-residue confidence $s_j \in [0, 1]$,

$$a_{m \to m}(j) = \max(s_j - \delta, \gamma), \qquad \delta = 0.05, \ \gamma = 0.5.$$

Profiles were searched with `hmmsearch` (HMMER3) [36, 37] against UniProtKB/TrEMBL (release 2025_03, 18 June 2025). As a single-sequence control, extracted (and reversed) chain sequences were searched with `phmmer` [38] against the same database.

Let $\mathcal{H} = \{h : E(h) < E_{\mathrm{cut}}\}$ with $E_{\mathrm{cut}} = 10^{-10}$. For query chain q,

$$h^\star(q) = \arg\min_{h \in \mathcal{H}_q} E(h),$$

where $\mathcal{H}_q \subseteq \mathcal{H}$ are hits to chain q. For each complex, the retained set is

$$\mathcal{R} = \mathrm{Top}_K\big(\mathcal{H}; E\big) \ \cup \ \big\{\, h^\star(q) : q \in \mathcal{Q} \,\big\}, \qquad K = 10,$$

the union of the global top-K hits by E-value and every per-chain best hit ($\mathcal{Q}$: chains of that complex; duplicates removed). Chain-level metrics use $h^\star(q)$; $\mathcal{R}$ records alternative matches.

4.4 Residue-type evaluation

Deposited and predicted Cα coordinates $\{r_i^{\mathrm{dep}}\}$ and $\{r_j^{\mathrm{pred}}\}$ were paired greedily under $d_{\mathrm{cut}} = 3$ Å (ascending distance; each site used at most once). For map m with N_m^{dep} deposited Cα sites and N_m^{corr} correctly typed paired sites (unpaired deposited sites count as incorrect),

$$\mathrm{Acc}_m = \frac{N_m^{\mathrm{corr}}}{N_m^{\mathrm{dep}}}.$$

Per-type recall for amino acid a uses the deposited count of type a as denominator. In Figure 2f, $\Delta_a = \mathrm{Recall}_a^{\mathrm{EModelS}} - \mathrm{Recall}_a^{\mathrm{EModelX}}$; solid points are pooled residue-level Δ_a, and bars are 95% bootstrap intervals of the unweighted per-map mean Δ_a.

For feature analysis, at most 80 patches (one per EMDB entry when available) were used. Within a patch, voxels at side-chain heavy-atom, non-Cα backbone (N/C/O) or Cα centres with standard amino-acid labels were eligible (at most 400 per patch). Features were L2-normalized with $\mathrm{sim}(u, v) = \langle u, v\rangle/(\|u\|\|v\|)$. Same-type pairs shared a residue label; different-type pairs did not. The t-SNE input was PCA-reduced to at most 30 dimensions; kNN used $k = 15$ with stratified five-fold cross-validation (voxels from the same map could occur in different folds).

4.5 Sequence-identification evaluation

The benchmark compares EModelS, EModelX (w/o seq), ModelAngelo and `findMySequence` on a common set $\mathcal{C}$ of $|\mathcal{C}| = 206$ chains from 99 complexes, formed as the union of chains scored by any method (outer join on complex and chain identifiers). Chains lacking a search hit for a given method are retained with SeqId = 0 rather than dropped from $\mathcal{C}$. EModelS and EModelX (w/o seq) infer models without a supplied target sequence. The `findMySequence` input protocol follows the existing benchmark setup.

Identified (query) and deposited (reference) sequences were aligned with EMBOSS `water` (Smith–Waterman; BLOSUM62 via EMBOSS `EBLOSUM62`, gap opening 10, extension 0.5) [39, 40]. We do not use the Identity percentage printed by EMBOSS (matches over alignment length). Instead, the pair-format alignment is re-parsed and metrics are recomputed with the full reference length as the identity denominator. For reference chain c, let $L_c = |c|$ and $|q_c|$ be the deposited and full query lengths. From the alignment strings, let M_c be the number of columns in which both sides are non-gap and identical, R_c the number of non-gap reference residues, and Q_c the number of non-gap query residues. Then

$$\mathrm{SeqId}_c = \frac{M_c}{L_c}, \qquad \mathrm{RefCov}_c = \frac{R_c}{L_c}, \qquad \mathrm{QryCov}_c = \frac{Q_c}{|q_c|}.$$

A chain-level hit is

$$\mathrm{Hit}_c = \mathbf{1}\big[\mathrm{SeqId}_c \geq \theta \ \wedge \ \mathrm{QryCov}_c \geq \theta\big], \qquad \theta = 0.90.$$

Chains with $\mathrm{Hit}_c = 0$ remain in the denominator and receive $\mathrm{SeqId}_c = 0$ in distributional plots. For $\mathcal{C}' \subseteq \mathcal{C}$, $\mathrm{HR}(\mathcal{C}') = |\mathcal{C}'|^{-1} \sum_{c\in\mathcal{C}'} \mathrm{Hit}_c$. At complex level, Fully matched means $\mathrm{Hit}_c = 1$ for every evaluated chain; Partially matched means a nonempty proper subset is a hit; complexes with no hit chain are omitted from the displayed categories.

Panels d–f report $\mathrm{HR}(\mathcal{C}')$ within strata. The Hard subset is

$$\mathcal{C}_{\mathrm{Hard}} = \big\{\, c : \mathrm{res}(c) \geq 3.5\,\text{Å} \ \wedge \ L_c > 400 \,\big\} \quad (|\mathcal{C}_{\mathrm{Hard}}| = 30),$$

and $\mathcal{C}_{\mathrm{Other}} = \mathcal{C} \setminus \mathcal{C}_{\mathrm{Hard}}$ $(|\mathcal{C}_{\mathrm{Other}}| = 176)$.

4.6 CryoSeek case studies

The three structural examples use sequence-free EModelS models from the CryoSeek application (legacy filename for the 9XFV Cα model: `9XFV_EModelX(noseq)_ca_model.pdb`). For 9XFV

panel b, TMalign on Cα atoms compares EModelS model chain W with deposited chain R; RMSD is taken from the aligned-length line and the TM-score is normalized by deposited chain length. For 9XNC and 9XNH, whole-chain local CC is Phenix `map_model_cc CC_mask` [29]. At aligned position i,

$$\Delta\mathrm{CC}_i = \mathrm{CC}_i^{\mathrm{dep}} - \mathrm{CC}_i^{\mathrm{search}},$$

the per-residue CC of the deposited side chain minus that of the searched sequence. Direction $X \to Y$ replaces searched residue X by deposited Y (and conversely for $Y \to X$). Only amino-acid pairs present in both directions with at least four total sites are shown. Each substitution model changes only the mismatched side chain, keeps the refined backbone, and is refined with `phenix.real_space_refine` [41] (five macro cycles, one process, map resolution) before recomputing CC. Deposited versus searched sequences are aligned with EMBOSS `water`; unmatched positions are not counted as substitutions.

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Supplementary Information

A Contrastive Learning–Based Sequence Identification Framework
for Bottom-Up Cryo-EM Structural Discovery

In sequence-identification results, **EModelS(s)** denotes direct sequence search of the modeled sequence with `phmmer`, and unmarked **EModelS** denotes HMM profile search with `hmmsearch`. The same `(s)` / unmarked convention is used for EModelX.

1 Supervised contrastive learning for residue-type embeddings

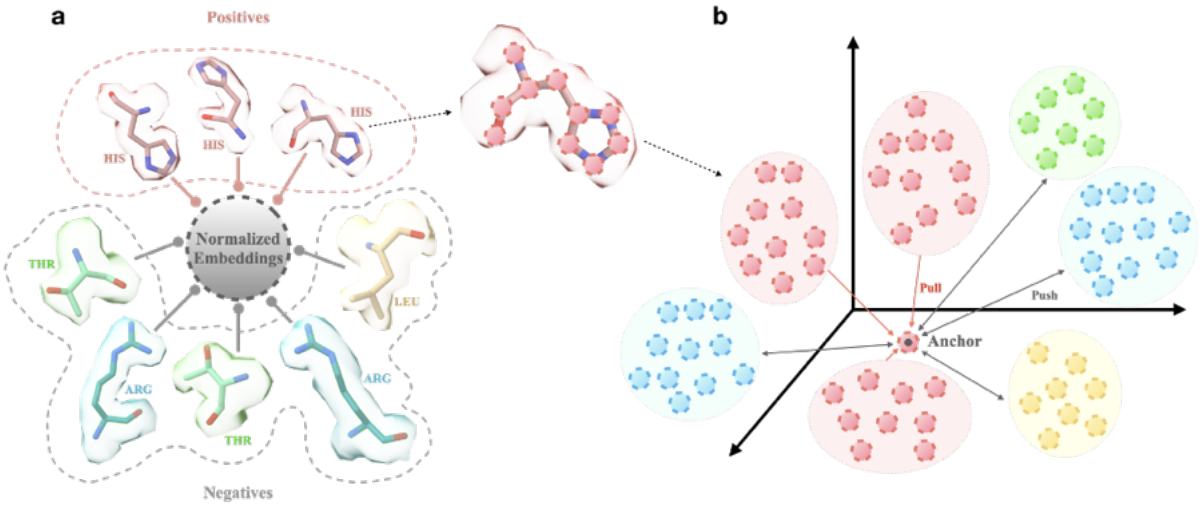


Figure 1: Geometric idea behind the contrastive term. Local density patches around atoms of the same amino-acid type are treated as positives and pulled together in the normalized embedding space, whereas patches of different types are treated as negatives and pushed apart. The same amino-acid type can adopt distinct side-chain conformations and density shapes in experimental maps.

2 Absolute per-amino-acid residue-type recall

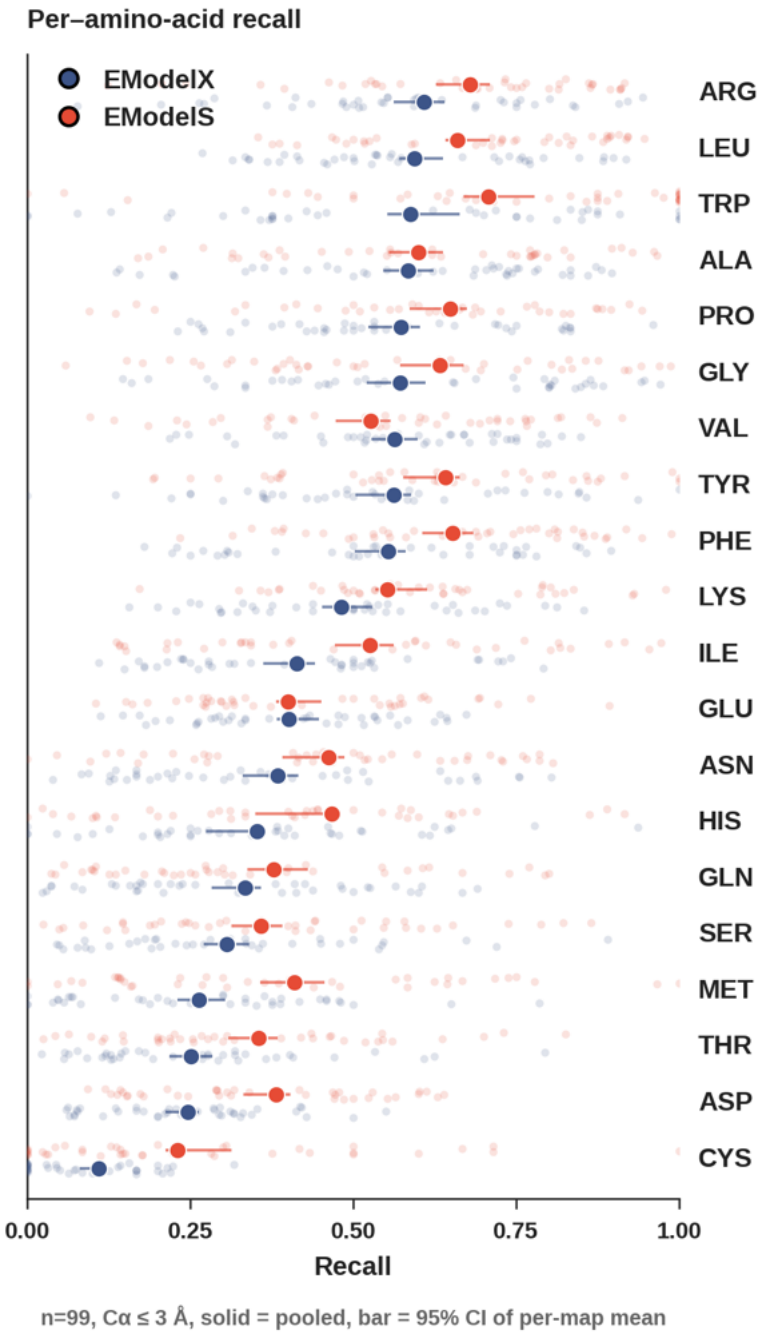


Figure 2: Absolute per-amino-acid residue-type recall for EModelX and EModelS on the 99-map benchmark, with a small vertical dodge per residue type. Solid points are pooled recalls (correct residues divided by deposited Cα counts of that type within 3 Å pairing). Horizontal bars are 95% bootstrap confidence intervals of the unweighted per-map mean. Amino-acid types are labelled on the right; residues are ordered by EModelX pooled recall (lowest at the bottom). Cysteine, aspartate and methionine are among the lowest EModelX recalls and also among the largest gains for EModelS.

3 Benchmark correlations of accuracy, resolution and chain length

On the 99-map / 206-chain benchmark, residue-type accuracy and sequence identity both decline as map resolution worsens, and higher residue-type accuracy is associated with higher SI (Figure 3). Overall, EModelS remains above EModelX across the resolution range for amino-acid accuracy, and above both EModelX and `findMySequence` for SI. The clearest gains for EModelS appear on long chains and on lower-resolution maps, consistent with the main-text analysis.

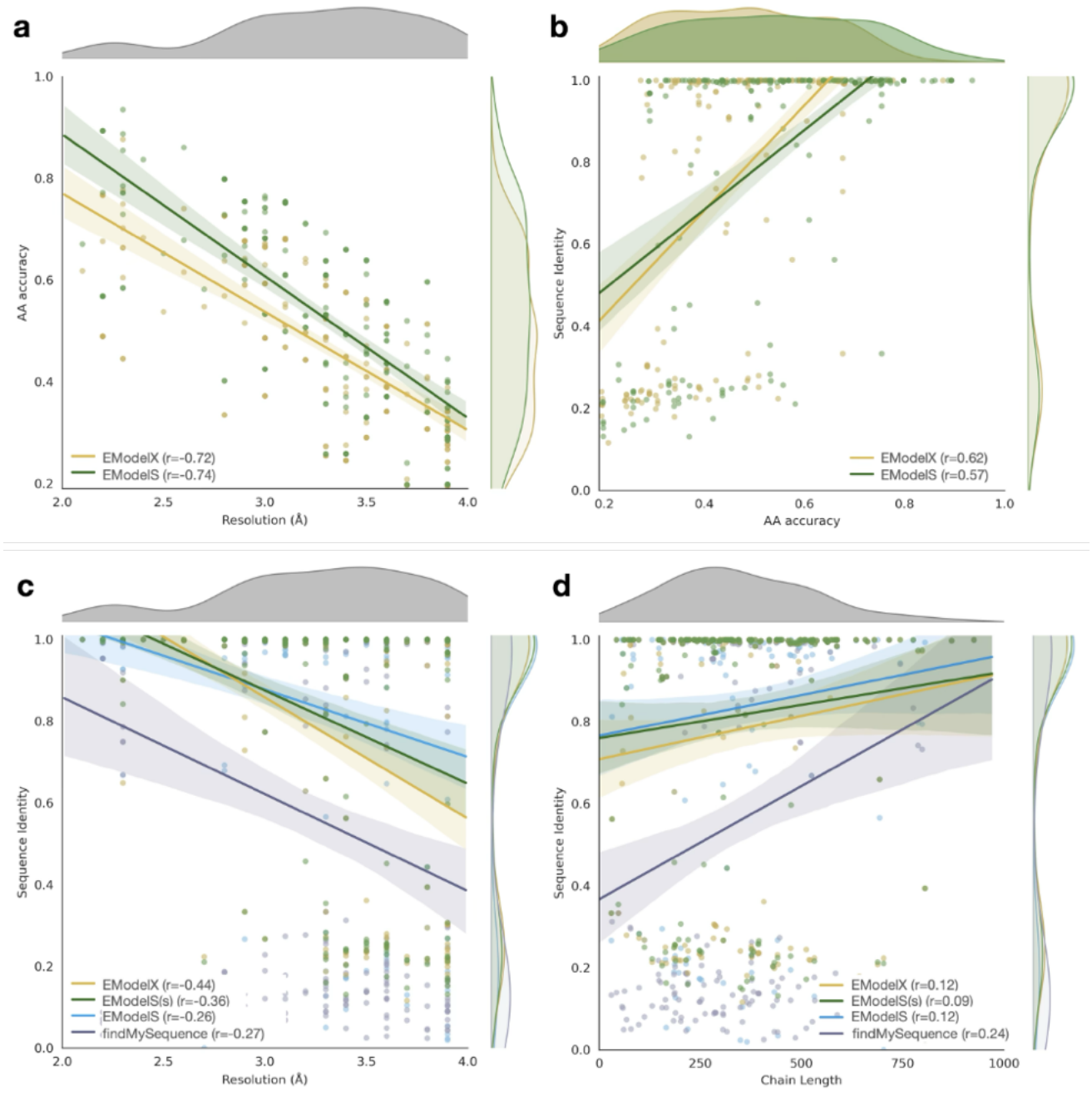


Figure 3: Benchmark correlations. **a,** Amino-acid accuracy versus map resolution for EModelX and EModelS. **b,** Sequence identity versus amino-acid accuracy for sequence search (`phmmer`; EModelX(s) / EModelS(s)). **c,** Sequence identity versus resolution for sequence search (EModelX(s) / EModelS(s)) and HMM profile search (EModelS), including `findMySequence`. **d,** Sequence identity versus deposited chain length under the same search modes as (**c**). Solid lines are linear fits; shaded bands are 95% confidence intervals; marginal densities are shown on the top and right.

4 Case study: needle filament–tip complex (PDB 7RYE)

The *Salmonella* type III secretion injectisome tip complex (PDB 7RYE, 3.9 Å) contains many short chains of three sequence types. Table 1 and Figure 4 reproduce the sequence-search outcome for representative chains A, E and F. A <1% gain in residue-type accuracy (0.31 to 0.32) was sufficient for EModelS(s) to recover UniProt

A0A379W4M3 for chain F (SI 0.962), whereas EModelX(s) returned a distant hit (SI 0.255). HMM profile search with EModelS and `findMySequence` retrieved longer N-terminally extended homologs with lower SI. The network-predicted sequences of EModelS modeled chains M, P (reversed) and S (reversed) retrieve A0A379W4M3 by `phmmer`, reflecting the repetitive tip architecture.

Table 1: Sequence identification for PDB 7RYE. Each cell lists the top UniProt accession and SI to the deposited chain.

Method	Chain A (78)	Chain E (77)	Chain F (212)
EModelX(s)	A0A5T3CYD6 : 0.987	A0A5T3CYD6 : 0.987	A0A410G1X1 : 0.255
EModelS(s)	A0A977R3G5 : 1.000	A0A977R3G5 : 1.000	A0A379W4M3 : 0.962
EModelS	A0A3Q9MPX2 : 1.000	A0A3Q9MPX2 : 1.000	A0A2H9U915 : 0.608
findMySequence	A0A5U9SPJ6 : 0.295	A0A5U9SPJ6 : 0.299	A0A3V2NZK6 : 0.774

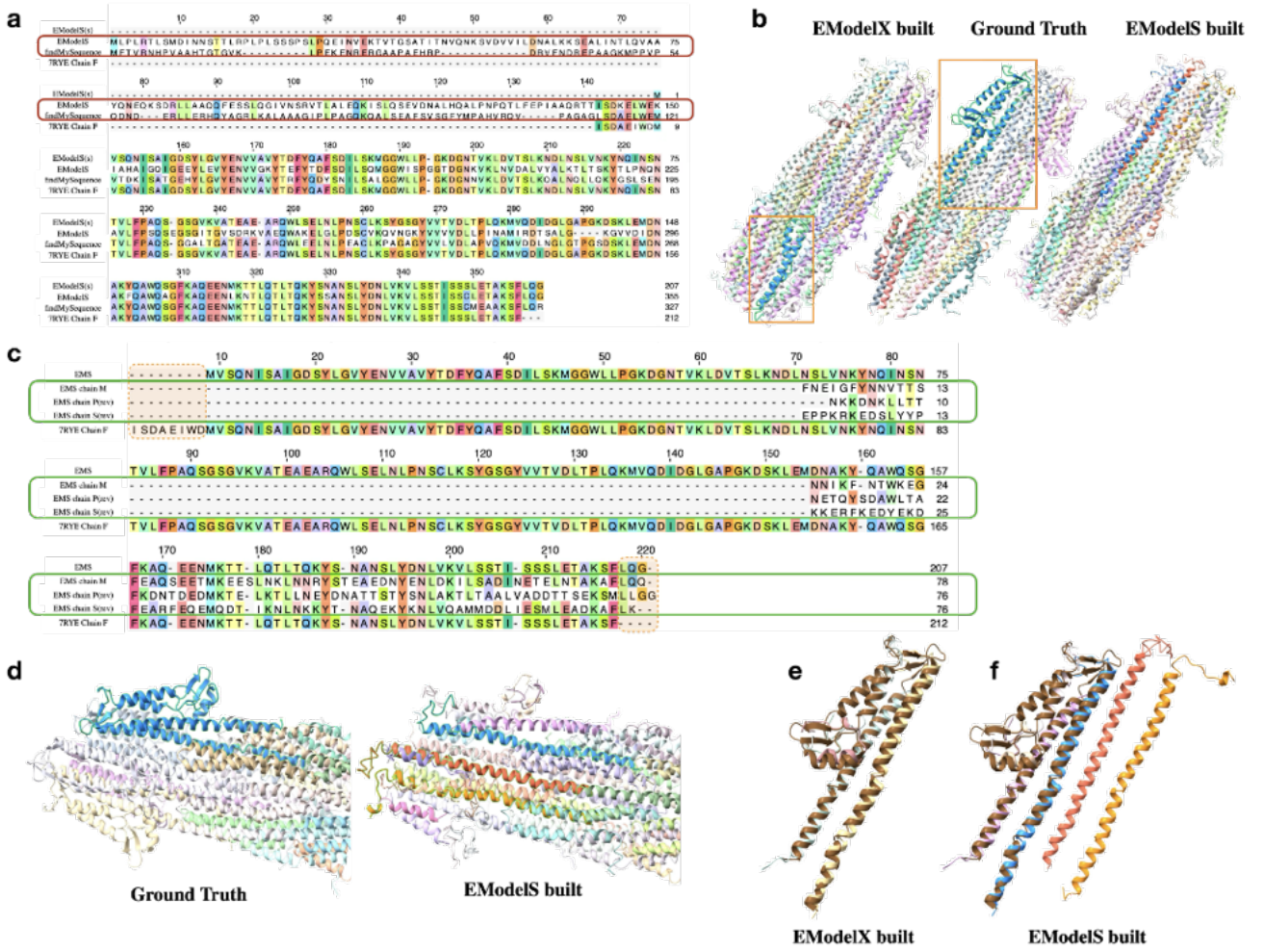


Figure 4: Sequence alignment and structural comparison for chain F of PDB 7RYE. **a,** Searched UniProt sequences versus the deposited chain F: the top hit from EModelS(s)/`phmmer` (A0A379W4M3; SI 0.962), from EModelS/`hmmsearch` (A0A2H9U915; SI 0.608) and from `findMySequence` (A0A3V2NZK6; SI 0.774). **b,** Overall tip-complex structures from EModelX, the deposited model and EModelS (left to right). **c,** Network-predicted sequences versus the deposited chain F for the EModelS modeled chains M, P (reversed) and S (reversed), which all retrieve A0A379W4M3 by `phmmer`. The unlabeled EModelS row is that shared searched UniProt sequence (A0A379W4M3). **d,** Local views of the tip region from panel **b**, showing deposited chain F and the neighbouring EModelS chains M, P and S. **e,f,** Individual chains extracted for comparison: deposited chain F superimposed with the corresponding modeled chains from EModelX and EModelS.

5 Per-chain sequence identification on the 99-map benchmark

On the 206 chains from the 99-map benchmark, sequence search (`phmmer`) of sequences read from sequence-free models gave a mean SI of 0.783 for EModelX(s) and 0.817 for EModelS(s). SI was at least 0.90 for 139/206 (EModelX(s)) and 152/206 (EModelS(s)) chains. The joint criterion SI ≥ 0.90 and query coverage ≥ 0.90 was met by 72/206 and 80/206 chains. Exact recovery (SI = 1) occurred for 83/206 and 94/206 chains. Figures 5 and 6 show the chain-wise comparison and the length dependence; Table 2 lists every chain.

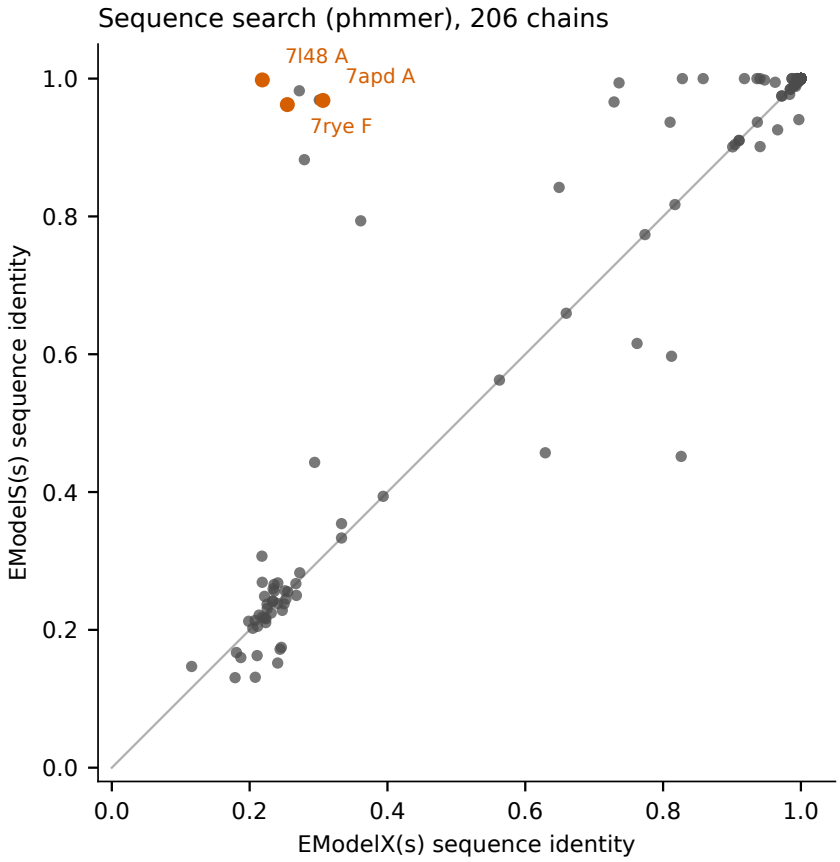


Figure 5: Sequence identity from sequence search (`phmmer`) of sequences read from sequence-free models (EModelX(s) versus EModelS(s)). Each point is one deposited chain. The gray line marks equal identity.

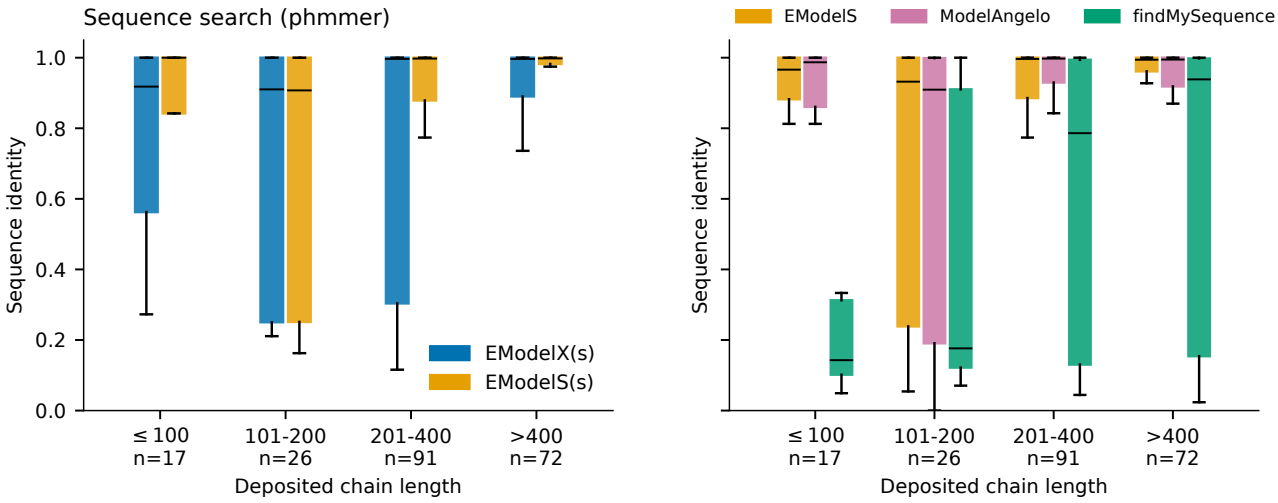


Figure 6: Sequence identity by deposited chain length. Boxes show the median and interquartile range; whiskers extend to 1.5 times the interquartile range. Left: sequence search (EModelX(s) / EModelS(s)). Right: HMM profile search (EModelS), together with ModelAngelo and `findMySequence`.

Table 2: Per-chain sequence identity on the 99-map benchmark. EModelX(s) and EModelS(s) are reference-length-normalized identities from sequence search with `phmmer`. Query coverage is the fraction of the identified sequence covered by the local alignment. EModelS, ModelAngelo, and findMySequence are identities from HMM profile search (`hmmsearch`) or backbone-based searches.

PDB	Chain	Len	EModelX(s)	EModelS(s)	EModelX(s) q	EModelS(s) q	EModelS	MA	fMS
7a4a	A	439	0.936	1.000	0.775	0.171	1.000	1.000	0.936
7akv	A	378	0.246	0.175	0.106	0.297	0.780	0.418	0.119
7akv	G	421	0.998	0.998	0.473	0.473	0.998	0.998	0.995
7alw	A	344	1.000	1.000	0.489	0.489	1.000	1.000	0.988
7apd	A	284	0.306	0.968	0.221	0.469	0.996	1.000	0.088
7apd	B	289	0.301	0.969	0.221	0.477	0.997	1.000	0.090
7apd	C	287	0.303	0.969	0.221	0.474	0.997	1.000	0.091
7apd	F	286	0.304	0.969	0.221	0.472	0.997	1.000	0.091
7apd	G	152	0.941	0.901	0.368	0.239	0.954	0.954	0.164
7apd	H	148	0.966	0.926	0.368	0.239	0.980	0.980	0.169
7b03	A	217	1.000	1.000	0.896	0.900	1.000	1.000	0.995
7b2l	A	147	1.000	1.000	0.742	0.742	1.000	1.000	0.218
7b2l	B	247	1.000	1.000	0.821	0.821	1.000	1.000	1.000
7b4l	A	443	1.000	1.000	0.825	0.825	0.995	0.998	0.998
7cxr	A	307	0.762	0.616	0.960	0.479	0.964	0.792	0.104
7cye	A	326	1.000	1.000	0.992	0.992	1.000	1.000	1.000
7d7f	A	446	1.000	1.000	0.547	0.547	1.000	1.000	0.274
7d7f	B	482	0.992	0.992	0.634	0.634	0.961	0.992	0.990
7d7f	C	474	0.992	0.992	0.624	0.624	0.962	0.992	0.989
7d7f	D	471	0.992	0.992	0.632	0.632	0.964	0.992	0.989
7d84	A	159	0.994	0.994	0.377	0.377	1.000	1.000	0.994
7dde	A	525	0.901	0.901	1.000	1.000	0.669	0.669	0.669
7deg	A	581	1.000	1.000	0.992	0.992	1.000	1.000	1.000
7deg	B	147	1.000	1.000	0.987	0.987	1.000	0.578	0.184
7deg	C	32	0.562	0.562	0.842	0.727	0.812	0.812	0.312
7dmp	A	186	0.629	0.457	0.536	0.267	0.629	0.909	0.134
7dmp	B	336	0.997	0.940	0.583	0.582	0.997	0.997	0.896
7dmp	C	242	0.996	0.996	1.000	1.000	0.996	0.880	0.277
7dsx	B	509	0.998	0.998	0.625	0.667	0.998	0.998	0.992
7dsx	D	185	0.222	0.249	0.204	0.049	0.189	0.189	0.184
7dxi	A	371	1.000	1.000	0.827	0.827	1.000	1.000	0.111
7e1y	A	538	0.998	0.998	0.840	0.840	0.998	0.998	0.998
7e4h	A	439	1.000	1.000	0.950	0.950	1.000	1.000	0.121
7e4h	B	317	1.000	1.000	0.964	0.964	1.000	0.997	0.997
7e4h	C	291	0.997	0.997	0.997	0.997	0.969	0.997	0.168
7e4h	D	307	0.267	0.267	0.621	0.621	0.163	0.173	0.085
7ed5	A	926	1.000	1.000	0.142	0.142	0.999	0.999	1.000
7ed5	B	150	1.000	1.000	0.063	0.063	1.000	1.000	1.000
7ed5	C	72	1.000	1.000	0.010	0.011	1.000	1.000	1.000
7ed5	D	154	1.000	1.000	0.065	0.065	1.000	1.000	1.000
7eep	A	529	1.000	1.000	0.985	0.985	1.000	1.000	1.000
7eep	M	182	1.000	1.000	1.000	1.000	1.000	1.000	0.154
7en0	A	313	1.000	1.000	0.558	0.558	1.000	1.000	1.000
7evp	A	377	1.000	1.000	1.000	1.000	1.000	1.000	0.854
7evp	C	43	1.000	1.000	0.581	0.581	1.000	1.000	0.140
7f0j	B	386	1.000	1.000	0.630	0.630	0.997	0.974	0.876
7f3x	A	449	1.000	1.000	0.896	0.896	1.000	1.000	1.000
7jzy	A	427	0.937	0.937	0.979	0.979	0.960	0.958	0.166
7jzy	B	306	1.000	1.000	0.979	0.979	0.647	1.000	0.049
7jzy	C	187	0.251	0.257	0.328	0.146	0.209	0.241	0.139
7jzy	D	298	0.997	0.997	0.980	0.980	0.997	0.997	0.990
7jzy	E	335	0.997	0.997	0.980	0.980	0.997	0.997	0.991
7jzy	H	330	0.997	0.997	0.980	0.980	0.997	0.997	0.991
7jzy	J	68	0.279	0.882	0.205	1.000	0.882	0.941	0.074
7knq	A	588	1.000	1.000	0.869	0.869	1.000	1.000	1.000
7kxr	A	562	1.000	1.000	0.765	0.765	1.000	1.000	1.000
7kxr	L	220	0.241	0.268	0.040	0.309	0.223	0.164	0.164
7l48	A	522	0.218	0.998	0.115	0.987	0.998	0.042	0.998
7l48	B	519	0.220	0.998	0.115	0.981	0.998	0.042	0.998
7l6q	A	311	1.000	1.000	0.907	0.907	1.000	1.000	1.000
7l7s	H	524	1.000	1.000	0.914	0.914	0.987	0.992	0.927
7lar	A	183	0.995	1.000	0.583	0.920	0.995	0.995	0.995
7lb5	A	260	1.000	1.000	0.841	0.841	1.000	1.000	0.788
7ld5	A	587	0.985	0.985	0.768	0.768	0.985	0.985	0.985
7lf6	A	407	0.361	0.794	0.821	0.867	0.993	0.870	0.025
7lg5	A	871	0.998	0.998	1.000	1.000	0.998	0.998	0.998
7lgu	A	680	1.000	1.000	0.958	0.958	0.997	0.946	0.984
7lhg	C	805	0.394	0.394	0.864	0.864	0.994	0.186	0.024
7lhg	D	215	0.219	0.219	0.074	0.199	0.209	1.000	0.047
7lhg	G	316	0.294	0.443	1.000	0.934	0.443	0.326	0.044
7lhg	K	156	0.212	0.205	0.200	0.340	0.295	0.109	0.103
7lhv	A	575	1.000	1.000	0.839	0.839	0.955	1.000	1.000
7lic	A	377	0.225	0.231	0.427	0.373	0.188	0.180	0.170
7lw1	A	744	1.000	1.000	0.954	0.954	0.944	0.988	0.941

PDB	Chain	Len	EModelX(s)	EModelS(s)	EModelX(s) q	EModelS(s) q	EModelS	MA	fMS
7lzh	A	798	0.972	0.975	0.868	0.878	0.794	0.999	0.733
7lzh	B	788	0.972	0.975	0.868	0.878	0.798	0.999	0.742
7mca	A	493	0.947	0.998	0.604	0.613	0.980	0.998	0.154
7mca	B	253	0.941	1.000	0.427	0.427	0.953	1.000	0.206
7mca	C	584	1.000	1.000	0.977	0.977	1.000	1.000	0.187
7mca	D	438	0.995	1.000	0.915	0.915	0.993	1.000	0.993
7mca	E	452	1.000	1.000	0.998	0.998	1.000	0.996	0.175
7mca	F	164	0.250	0.238	0.159	0.073	0.220	0.195	0.122
7mca	I	376	0.253	0.245	0.395	0.916	0.231	0.215	0.213
7mge	A	349	0.241	0.152	0.109	0.383	0.023	0.046	0.103
7mge	B	442	0.208	0.131	0.117	0.247	0.034	0.100	0.154
7mge	C	365	0.181	0.167	0.246	0.275	0.019	0.055	0.115
7mge	E	166	0.211	0.163	0.037	0.224	0.054	0.127	0.102
7msw	A	635	1.000	1.000	0.144	0.144	0.998	0.998	1.000
7mxy	A	283	1.000	1.000	0.937	0.937	1.000	1.000	0.954
7mxy	B	281	1.000	1.000	0.930	0.930	1.000	1.000	0.954
7mxy	C	279	1.000	1.000	0.924	0.924	1.000	1.000	0.953
7mxy	D	282	1.000	1.000	0.934	0.934	1.000	1.000	0.954
7mxy	E	281	1.000	1.000	0.930	0.930	1.000	1.000	0.954
7n00	A	346	1.000	1.000	0.278	0.214	1.000	1.000	0.786
7n00	B	57	0.649	0.842	0.667	0.418	1.000	0.982	0.105
7n06	A	347	0.994	0.994	0.053	0.049	0.994	0.994	0.991
7n0b	A	131	0.214	0.221	0.070	0.092	1.000	0.000	0.107
7n0b	B	512	0.244	0.172	0.600	1.000	0.959	0.000	0.053
7n5d	A	386	0.984	0.984	0.753	0.753	0.679	0.984	0.984
7n5d	B	371	1.000	1.000	0.753	0.753	0.693	1.000	1.000
7n6q	A	375	0.963	0.995	0.882	0.770	0.995	0.995	0.987
7n88	A	532	0.224	0.216	0.039	0.583	0.130	0.000	0.177
7n88	B	688	0.208	0.214	0.196	0.236	1.000	0.000	0.093
7n8n	A	197	0.218	0.269	0.141	0.028	0.188	0.000	0.122
7n8n	B	202	0.218	0.307	0.067	0.387	0.248	0.000	0.069
7n98	A	473	0.998	0.998	0.886	0.886	0.998	0.998	0.896
7n9z	F	658	1.000	1.000	0.992	0.992	1.000	0.956	0.032
7n9z	G	262	1.000	1.000	0.888	0.888	0.992	0.992	0.088
7n9z	H	184	1.000	1.000	0.902	0.902	0.995	0.995	0.071
7n9z	I	98	1.000	1.000	0.899	0.899	1.000	1.000	0.061
7ng9	A	421	1.000	1.000	0.866	0.866	1.000	1.000	0.881
7nh9	A	327	1.000	1.000	0.985	0.985	1.000	1.000	1.000
7nhr	A	663	0.995	0.995	0.978	0.978	0.995	0.995	0.995
7nhr	B	564	0.995	0.995	0.978	0.978	0.995	0.995	0.995
7nhr	C	565	0.995	0.995	0.978	0.978	0.995	0.995	0.995
7nhr	E	525	0.994	0.994	0.978	0.978	0.994	0.994	0.994
7nk9	G	30	0.333	0.333	0.081	0.049	0.267	0.267	0.133
7nk9	H	79	0.810	0.937	0.847	0.975	0.886	0.861	0.051
7nk9	L	84	1.000	1.000	0.977	0.977	0.964	1.000	1.000
7nk9	O	85	1.000	1.000	0.988	0.988	0.953	1.000	0.988
7nk9	a	238	1.000	1.000	0.944	0.944	1.000	1.000	0.071
7nk9	b	61	0.918	1.000	0.370	0.359	1.000	1.000	0.049
7nk9	d	59	0.729	0.966	0.133	0.133	0.966	0.966	0.102
7np3	A	490	0.990	0.990	0.825	0.825	0.988	0.988	0.973
7nt5	A	394	0.236	0.256	0.204	0.822	0.995	0.997	0.137
7nt5	B	395	0.235	0.266	0.204	0.191	0.995	0.997	0.137
7nt5	L	393	0.234	0.260	0.205	0.822	0.995	0.997	0.137
7nt5	M	382	0.241	0.238	0.204	0.311	0.995	0.997	0.141
7nv0	A	430	0.205	0.202	0.044	0.582	0.205	0.195	0.107
7nv0	B	248	0.234	0.242	0.082	0.588	0.173	0.625	0.117
7nv0	C	253	0.225	0.237	0.082	0.588	0.170	0.621	0.123
7nv0	D	249	0.233	0.241	0.082	0.588	0.173	0.622	0.116
7nvh	A	717	1.000	1.000	0.797	0.797	1.000	1.000	1.000
7o6e	A	177	1.000	1.000	0.978	0.978	1.000	1.000	1.000
7odf	A	703	0.223	0.211	0.312	0.451	–	–	0.048
7og5	H	157	0.904	0.904	0.917	0.917	0.904	0.904	0.904
7og5	J	167	0.910	0.910	0.917	0.917	0.910	0.910	0.910
7og5	L	156	0.910	0.910	0.917	0.917	0.910	0.910	0.910
7ogp	A	361	0.817	0.817	0.950	0.950	0.817	0.817	0.127
7ogp	B	203	1.000	1.000	0.418	0.418	1.000	0.995	0.079
7ogp	C	693	0.659	0.659	0.986	0.986	0.566	0.566	0.102
7ogp	D	510	1.000	1.000	0.999	0.999	1.000	1.000	1.000
7ogp	E	377	1.000	1.000	0.998	0.996	1.000	1.000	0.263
7oj5	A	184	1.000	1.000	0.669	0.669	1.000	1.000	1.000
7onj	A	262	1.000	1.000	0.916	0.916	1.000	1.000	1.000
7ose	A	503	0.998	0.998	0.979	0.979	0.992	0.992	0.994
7ose	B	377	1.000	1.000	0.997	0.997	1.000	1.000	0.228
7osl	A	326	1.000	1.000	0.483	0.483	1.000	1.000	1.000
7osl	B	327	1.000	1.000	0.484	0.484	1.000	1.000	1.000
7ouf	D	258	0.996	0.996	0.310	0.310	0.996	0.996	0.996
7ouf	E	278	0.996	0.996	0.310	0.310	0.996	0.996	0.996
7ouf	F	296	0.774	0.774	0.326	0.326	0.774	0.196	0.189
7ovb	A	473	0.247	0.228	0.432	0.455	0.195	0.175	0.121
7ovb	B	259	0.826	0.452	0.647	1.000	1.000	0.923	0.158
7ovb	C	202	0.995	1.000	0.971	0.971	0.995	1.000	0.911

PDB	Chain	Len	EModelX(s)	EModelS(s)	EModelX(s) q	EModelS(s) q	EModelS	MA	fMS
7ovb	D	283	0.272	0.982	0.047	0.963	0.947	1.000	0.972
7ovb	E	188	0.255	0.255	0.393	0.218	0.229	0.234	0.202
7oxo	A	776	1.000	1.000	0.978	0.978	0.999	0.999	0.834
7p54	A	397	0.995	0.995	0.770	0.770	0.995	0.995	0.982
7pkb	E	206	0.990	0.990	0.920	0.920	0.990	0.990	0.990
7pko	A	313	0.994	1.000	0.987	0.987	0.914	1.000	1.000
7ppj	A	643	0.179	0.131	0.202	0.679	0.095	0.969	0.048
7ptu	A	474	0.812	0.597	0.570	0.943	1.000	0.755	0.812
7puy	A	219	0.187	0.160	0.049	0.019	0.050	0.000	0.160
7puy	a	189	0.222	0.217	0.037	0.491	0.127	0.000	0.090
7pxa	0	215	1.000	1.000	0.915	0.915	1.000	1.000	0.186
7pxa	H	222	1.000	1.000	0.763	0.763	1.000	1.000	1.000
7pxa	I	214	1.000	1.000	0.915	0.915	1.000	1.000	0.187
7pxa	J	223	1.000	1.000	0.766	0.766	1.000	1.000	1.000
7pxa	K	217	1.000	1.000	0.915	0.915	1.000	1.000	0.184
7pxa	U	234	1.000	1.000	0.804	0.804	1.000	1.000	1.000
7q6z	A	485	0.736	0.994	0.932	1.000	0.992	0.992	0.029
7q6z	B	112	0.268	0.250	0.036	0.173	0.268	0.071	0.080
7qla	A	433	0.199	0.212	0.021	0.285	0.141	1.000	0.044
7qla	C	441	0.231	0.224	0.100	0.302	0.268	0.213	0.043
7rak	U	507	1.000	1.000	0.934	0.934	1.000	1.000	0.998
7rkh	D	559	1.000	1.000	0.988	0.988	1.000	1.000	1.000
7rsl	A	235	1.000	1.000	0.870	0.870	1.000	1.000	1.000
7rtm	A	571	0.858	1.000	0.458	0.551	0.970	1.000	0.970
7ru9	A	306	0.984	0.977	0.950	0.985	0.958	0.987	0.088
7ru9	C	271	0.993	0.993	0.831	0.877	0.993	0.993	0.993
7ru9	D	99	0.273	0.283	0.134	0.147	0.283	0.263	0.152
7ru9	E	48	0.333	0.354	0.089	0.062	0.333	0.250	0.333
7rye	A	78	0.987	1.000	0.975	0.975	1.000	0.987	0.295
7rye	E	77	0.987	1.000	0.962	0.962	1.000	0.987	0.299
7rye	F	212	0.255	0.962	0.073	0.986	0.608	0.915	0.774
7sii	A	324	1.000	1.000	0.879	0.879	1.000	1.000	1.000
7sii	B	319	1.000	1.000	0.879	0.879	1.000	1.000	1.000
7sq6	A	467	0.998	0.998	0.841	0.841	0.998	0.998	0.998
7vd9	A	498	1.000	1.000	0.945	0.945	0.928	0.918	0.749
7vd9	D	497	1.000	1.000	0.943	0.943	0.928	0.920	0.751
7vda	A	496	0.992	0.992	0.965	0.965	0.982	0.992	0.919
7vdc	A	343	0.828	1.000	0.482	0.819	0.828	0.843	0.840
7vfe	A	529	1.000	1.000	0.966	0.966	1.000	1.000	0.998
7vhq	A	63	1.000	1.000	0.548	0.548	1.000	1.000	0.143
7vhq	E	267	1.000	1.000	0.746	0.746	1.000	0.996	0.255
7vhq	G	299	1.000	1.000	0.985	0.985	0.943	0.936	1.000
7vlx	Y	247	1.000	1.000	0.922	0.922	1.000	0.874	1.000
7vlx	Z	298	0.997	0.997	0.983	0.983	0.973	0.997	0.178
7voj	A	400	0.998	0.998	0.822	0.822	0.998	0.998	0.998
7vq2	A	354	0.116	0.147	0.077	0.206	0.105	0.000	0.068
7w72	A	588	1.000	1.000	0.987	0.987	0.983	0.772	0.085
7w72	K	330	1.000	1.000	0.918	0.918	0.858	0.742	0.133
7w72	S	533	0.992	0.989	0.963	0.960	0.992	0.567	0.120
7w72	T	527	0.998	0.998	0.910	0.910	0.964	0.533	0.068
7w72	U	420	1.000	1.000	0.966	0.966	0.976	1.000	1.000

6 Modeled chain-length distributions for EModelS and ModelAngelo

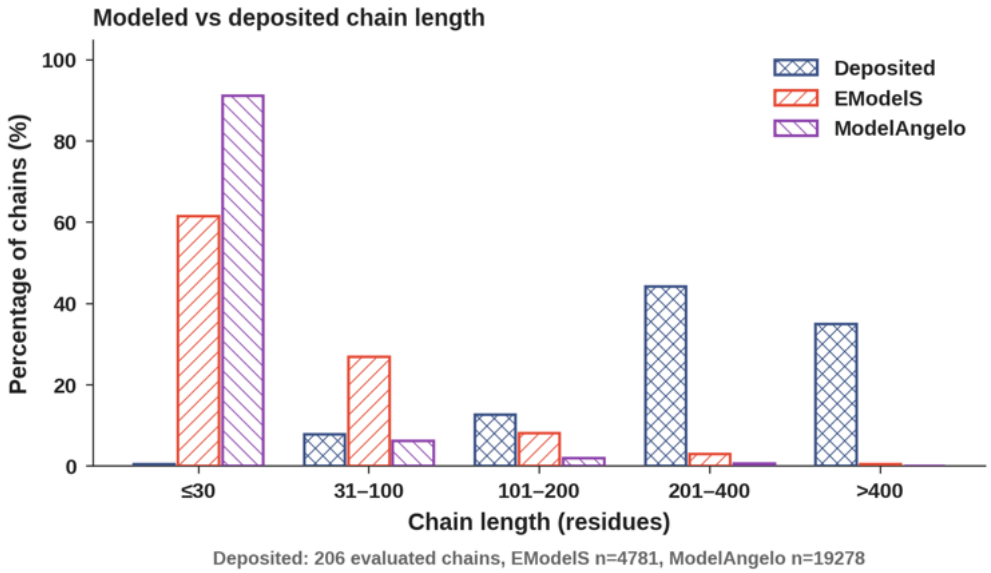


Figure 7: Chain-length distributions on the 99-map benchmark. Bars show the percentage of chains in each length bin for deposited reference chains ($n = 206$ evaluated chains) versus all modeled Cα chains from sequence-free EModelS ($n = 4781$) and ModelAngelo ($n = 19278$). Sequence search uses the automatically built chain as the query, so the length distribution of modeled chains affects how much contiguous sequence evidence is available. Both automatic methods are strongly enriched for short chains relative to the deposited distribution, consistent with long deposited chains being split into fragments; ModelAngelo places a larger share of chains in the shortest bin than EModelS.

7 Bacterial pili without a supplied sequence

Three pilus maps from a natural-source cryo-EM study were rebuilt with the same sequence-free protocol: pilus-like-α (PDB 9XFV, EMD-66838, 3.07 Å), pilus-like-β (9XNC, EMD-67046, 3.37 Å), and pilus-like-γ (9XNH, EMD-67051, 3.66 Å). No sequence was provided to either method. Both methods returned the same top UniProt entry for each map (Table 3); EModelS had the smaller E-value in every case. Residue-type accuracy on paired Cα atoms (3 Å, one-to-one) was 52.3% versus 57.8% (9XFV), 39.3% versus 45.6% (9XNC), and 40.7% versus 42.2% (9XNH) for EModelX versus EModelS. Figures 8–9 show per-position agreement on the best chain and the identity distribution across all traced chains of length at least 30. Residue-level alignments of the EModelS network sequence, the searched UniProt sequence and the deposited sequence are given in Figures 10–12.

Table 3: Top database hit for the three pilus maps. Trace identity is the local-alignment identity of the best modeled chain against the deposited sequence (EModelX / EModelS). Database identity is the number of identical residues in the local alignment of the retrieved sequence to the deposited chain, divided by the deposited chain length.

Map	Accession	E-value, EModelX	E-value, EModelS	Trace identity	Database identity
9XFV	A0A257CJ09	2.2×10^{-56}	8.6×10^{-59}	0.601 / 0.618	0.983
9XNC	A0A4Q7Z3W7	2.3×10^{-31}	2.2×10^{-40}	0.444 / 0.491	0.602
9XNH	A0A0D7KGF4	8.8×10^{-25}	2.9×10^{-32}	0.429 / 0.462	0.457

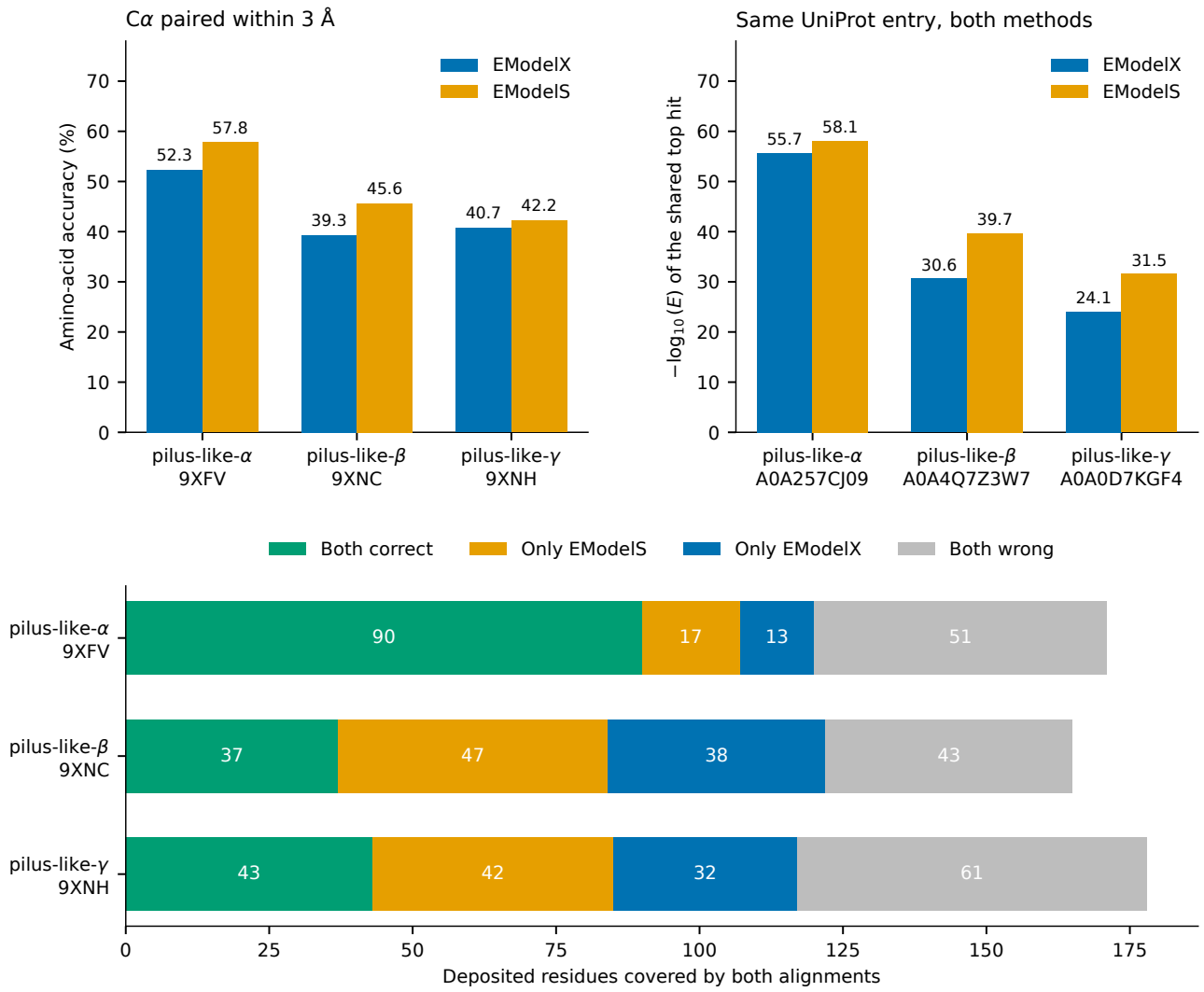


Figure 8: Pilus maps rebuilt without a supplied sequence. Top left: residue-type accuracy. Top right: $-\log_{10}(E)$ of the shared top hit. Bottom: deposited positions covered by both alignments on the best-matching chain, split by which method matched the deposited residue.

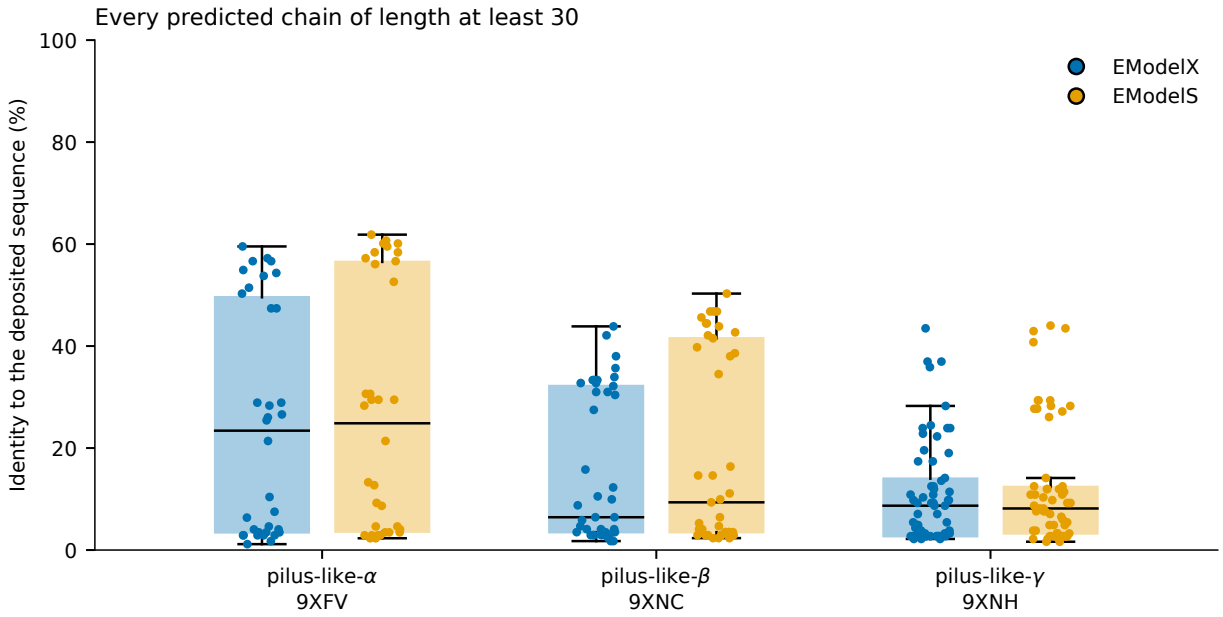


Figure 9: Identity of every modeled chain of length at least 30 to the deposited pilus sequence. Points are individual chains. The box shows the median and interquartile range.

Three-track sequence alignments for the pilus cases

For the EModelS-selected chain of each pilus map, Figures 10–12 align three sequences column by column on the residue numbering of the refined searched model: the network-read sequence (EModelS), the UniProt sequence returned by database search (Searched), and the deposited sequence (Deposited). A hyphen on the deposited row means that position has no deposited residue under the local alignment. Coloured cells mark disagreements with the deposited residue when it is present; where the deposited row is empty, Searched and EModelS are coloured when they disagree with each other.

9XFV (EMD-66838). The searched model has 207 residues, of which 170 align to the deposited chain (173 residues). On those paired positions the searched UniProt sequence is identical to the deposit (170/170); reference-normalized database identity is therefore $170/173 = 0.983$, and this case has no deposited-versus-searched mismatches for the later CC analysis. The EModelS network sequence matches the deposit at 104/170 positions (61%). The N-terminal block without a deposited assignment shows additional EModelS–Searched disagreements that are not counted in the paired identity.

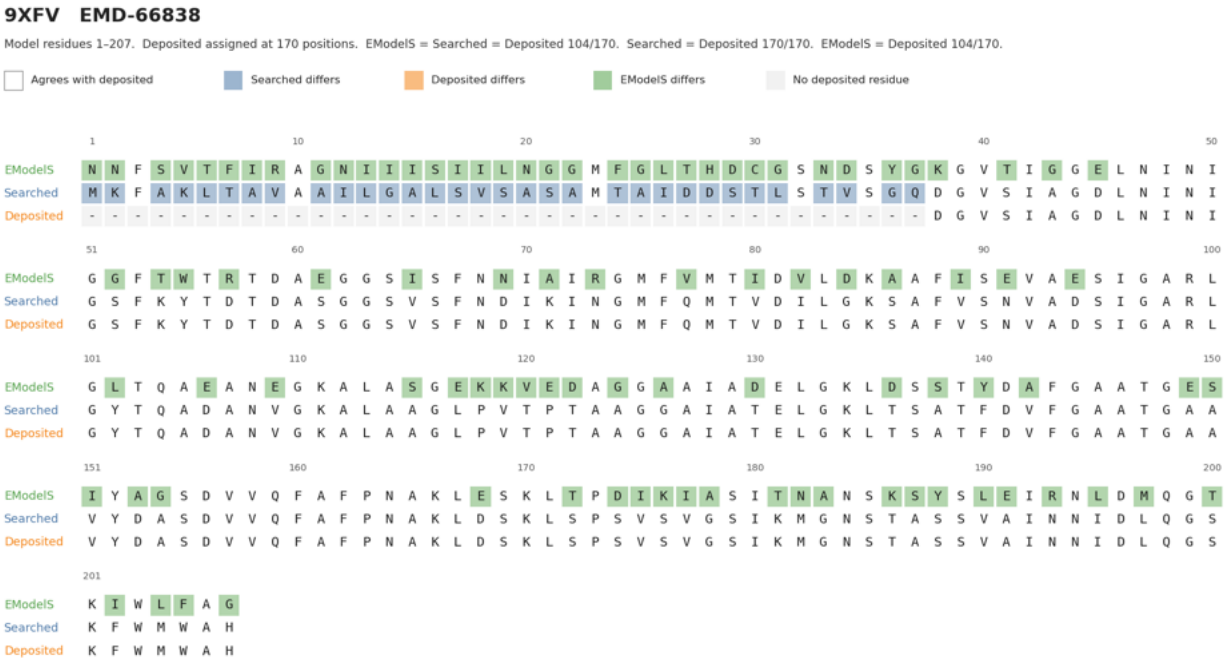


Figure 10: Three-track sequence alignment for pilus-like-α (PDB 9XFV, EMD-66838). Rows are EModelS, Searched, and Deposited. White cells agree with the deposited residue; green / blue / orange mark disagreements as in the legend; gray marks positions without a deposited residue.

9XNC (EMD-67046). All 163 model residues are paired to the deposited chain. The searched sequence matches at 97/163 positions (60%), and EModelS matches at 73/163 (45%); 57 positions are identical in all three tracks. Unlike 9XFV, blue/orange highlights appear throughout the chain where the PilA homolog retrieved by search differs from the deposit—these are the mismatch sites used in Figure 13.

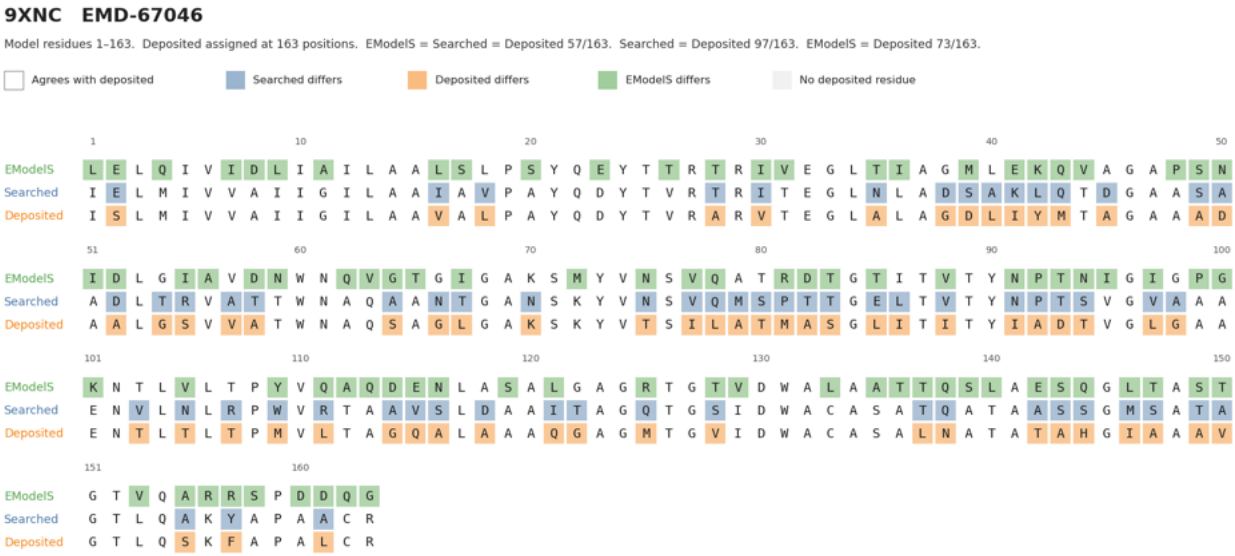


Figure 11: Three-track sequence alignment for pilus-like-β (PDB 9XNC, EMD-67046). Notation as in Figure 10.

9XNH (EMD-67051). The searched model has 173 residues, with 160 paired to the deposit. Searched

matches at 82/160 (51%) and EModelS at 62/160 (39%); 42 positions agree in all three tracks. As for 9XNC, remaining paired disagreements between Searched and Deposited define the mismatch set for the CC comparison in Figure 13.



Figure 12: Three-track sequence alignment for pilus-like-γ (PDB 9XNH, EMD-67051). Notation as in Figure 10.

8 Mismatch-site map–model CC after sequence search

After sequence-free EModelS modeling and UniProt search on the pilus maps (PDB 9XNC and 9XNH; 9XFV has no deposited-versus-searched residue mismatches under the alignment used here), each mismatch site was rebuilt with the searched amino acid or the deposited amino acid and rescored by real-space CC. Figure 13 summarizes the 143 mismatch sites; the main text reports the bidirectional substitution summary for these cases.

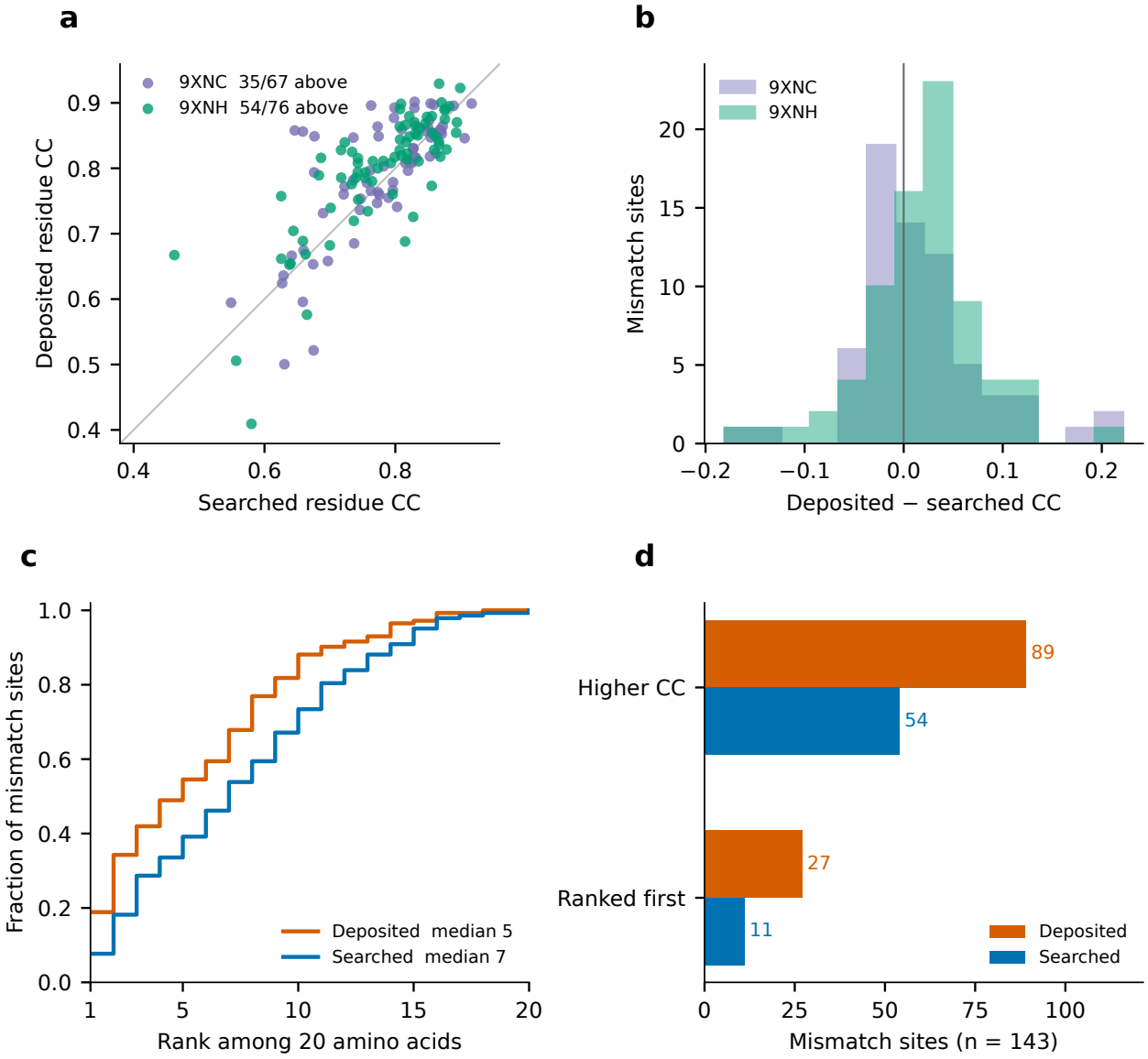


Figure 13: Mismatch-site CC after sequence search. **a,** Deposited versus searched residue CC; points above the diagonal favour the deposited amino acid. **b,** Distribution of deposited−searched CC. **c,** Cumulative rank of deposited and searched amino acids among all 20 types by CC. **d,** Counts of sites with higher CC or rank 1 for deposited versus searched residues.

On these mismatch sites the deposited amino acid more often has the higher CC (89 versus 54) and more often ranks first among the 20 amino acids (27 versus 11). The mean CC difference is modest (+0.0175), so the signal is statistical rather than a large per-residue margin.

9 Case study: NDCBE / SLC4A8 (PDB 7RTM)

When the true sequence is supplied for atomic building, higher residue-type accuracy improves chain assignment. For the sodium-driven bicarbonate exchanger NDCBE (Figure 14), EModelS raised amino-acid accuracy from 0.39 to 0.44 relative to EModelX. Two terminal regions that control chain continuity are shown at higher magnification: improved local residue-type predictions correct junction geometry and prevent the A/B register swaps seen with EModelX. This case shows that higher residue-type accuracy improves chain connectivity even when sequence identity is not the limiting step.

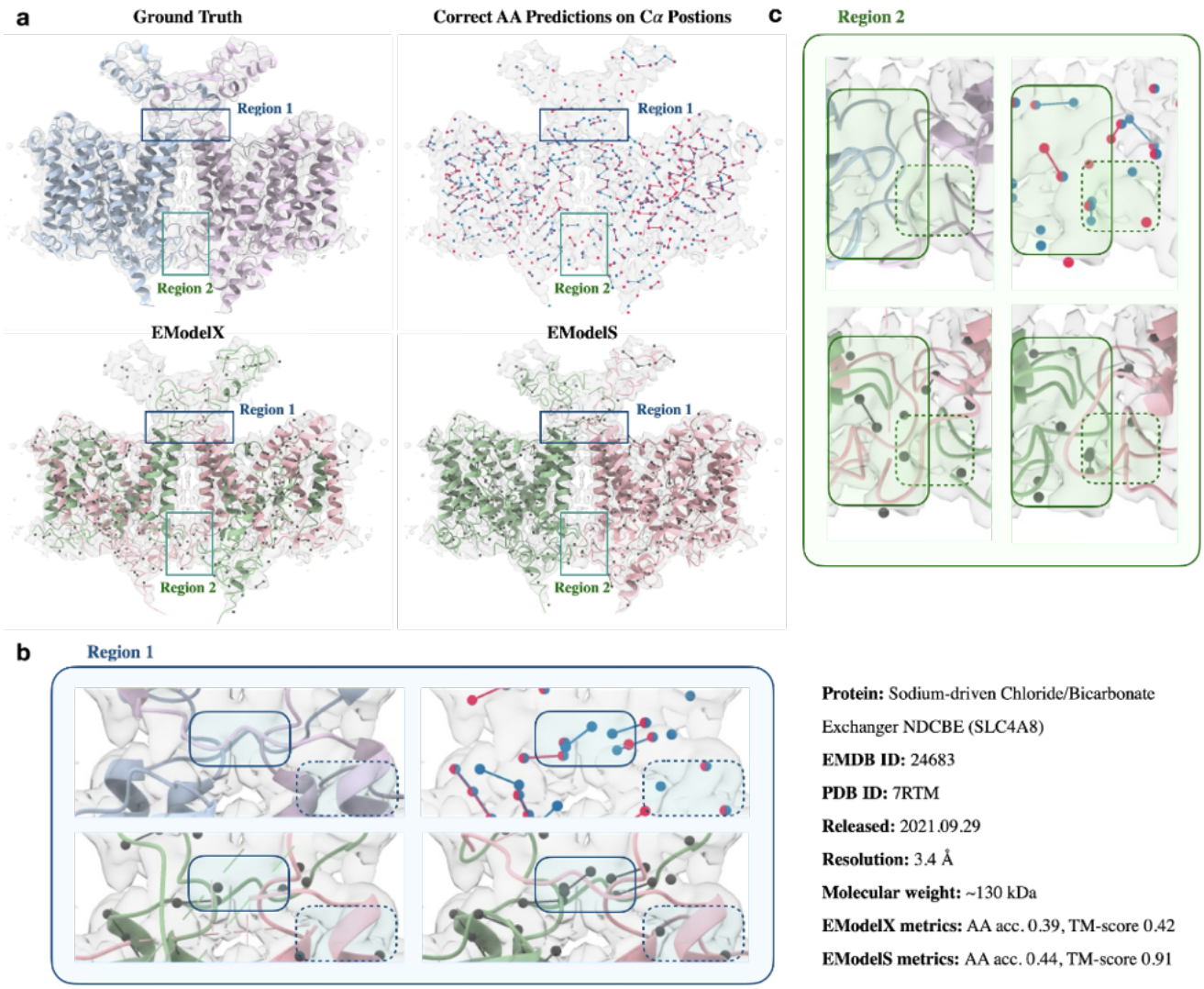


Figure 14: Predicted models of NDCBE (SLC4A8) compared with the deposited structure. **a,** Overall models fitted in the experimental map and coloured by chain: deposited structure with Cα sites of correct residue-type prediction marked for EModelX (red) and EModelS (blue), together with the EModelX and EModelS builds; two terminal regions that control chain continuity are enlarged in panels **b** and **c**. **b,** Magnified view of region 1, where improved local residue-type prediction restores the correct chain junction and prevents the A/B register swap seen with EModelX. **c,** Magnified view of region 2 at the opposite terminus, where the same local improvement recovers continuous chain assignment.

10 Contrastive training ablations

Table 4 reports per-voxel amino-acid classification accuracy under alternative training regimes on the full training set and a 1% subset. Loss names follow the main text: $\mathcal{L}_{\text{CE}}$ is the label-smoothed cross-entropy used for dense residue-type classification, and $\mathcal{L}_{\text{sup}}$ is the InfoNCE-based supervised contrastive loss on atom-centred embeddings; the EModelS objective is $\mathcal{L} = \mathcal{L}_{\text{CE}} + \overline{\mathcal{L}_{\text{sup}}}$. MSE and cosine denote experimental–simulated feature-alignment losses: the same network embeds a cryo-EM patch and its simulated density counterpart, and the two feature maps are pulled together by an MSE (or SmoothL1/SSIM) or cosine-distance term without using residue-type labels. "Cα" versus "all atoms" controls which sites enter $\mathcal{L}_{\text{sup}}$; rows 4.x–5.1 use stagewise training ($\mathcal{L}_{\text{sup}}$ pretraining, optionally with MSE alignment, freeze the encoder, then train a $\mathcal{L}_{\text{CE}}$ head), whereas the remaining rows optimize the listed losses jointly. Dashes mark configurations that were not evaluated.

On the full training set, adding an experimental–simulated alignment term to $\mathcal{L}_{\text{CE}}$ does not help: $\mathcal{L}_{\text{CE}}$+MSE

(1.1) matches the $\mathcal{L}_{\mathrm{CE}}$-only benchmark (0.503 versus 0.504), and $\mathcal{L}_{\mathrm{CE}}$+cosine (1.2) is slightly worse (0.496). Replacing that term with supervised $\mathcal{L}_{\mathrm{sup}}$ is sensitive to the sampling sites: $\mathcal{L}_{\mathrm{CE}} + \mathcal{L}_{\mathrm{sup}}$ on Cα centres only (2.1) remains near the benchmark (0.498), whereas the same joint objective on all atomic centres (2.2) reaches 0.512, the best full-set result and the setting retained for EModelS. Adding MSE alignment on top of $\mathcal{L}_{\mathrm{CE}} + \mathcal{L}_{\mathrm{sup}}$ at Cα sites (3.1; 0.499) gives no further gain over 2.1, so the alignment term was not kept in the final objective. Under stagewise training, freezing after $\mathcal{L}_{\mathrm{sup}}$ on Cα sites (4.1) collapses accuracy to 0.376, whereas freezing after $\mathcal{L}_{\mathrm{sup}}$ on all atoms (4.2) recovers 0.507, close to joint all-atom training (2.2) and to $\mathcal{L}_{\mathrm{CE}}$ alone; thus the all-atom versus Cα contrast persists when the encoder is frozen, but joint $\mathcal{L}_{\mathrm{CE}} + \mathcal{L}_{\mathrm{sup}}$ (2.2) remains preferable to the two-stage protocol.

On the 1% subset, $\mathcal{L}_{\mathrm{CE}}$ alone reaches 0.307. Stagewise training remains below this baseline for every evaluated freeze protocol: 0.144 after $\mathcal{L}_{\mathrm{sup}}$ on Cα (4.1), 0.283 after $\mathcal{L}_{\mathrm{sup}}$ on all atoms (4.2), and 0.170 after MSE+$\mathcal{L}_{\mathrm{sup}}$ on Cα (5.1). Relative to 4.1, adding MSE alignment during Cα pretraining (5.1) helps modestly, and expanding $\mathcal{L}_{\mathrm{sup}}$ pretraining from Cα to all atoms (4.2 versus 4.1) helps more, yet neither closes the gap to $\mathcal{L}_{\mathrm{CE}}$-only training when data are scarce. Overall, the controlled comparisons favour joint $\mathcal{L}_{\mathrm{CE}} + \mathcal{L}_{\mathrm{sup}}$ on all atomic centres for the full set, while experimental–simulated alignment losses and Cα-restricted $\mathcal{L}_{\mathrm{sup}}$ sampling do not improve over $\mathcal{L}_{\mathrm{CE}}$, and stagewise freeze protocols underperform especially on the small subset.

Method	Full set	1% subset
1.1 $\mathcal{L}_{\mathrm{CE}}$ + MSE	0.503	–
1.2 $\mathcal{L}_{\mathrm{CE}}$ + cosine	0.496	–
2.1 $\mathcal{L}_{\mathrm{CE}}$ + $\mathcal{L}_{\mathrm{sup}}$ (Cα)	0.498	–
2.2 $\mathcal{L}_{\mathrm{CE}}$ + $\mathcal{L}_{\mathrm{sup}}$ (all atoms)	0.512	–
3.1 $\mathcal{L}_{\mathrm{CE}}$ + MSE + $\mathcal{L}_{\mathrm{sup}}$ (Cα)	0.499	–
4.1 Freeze after $\mathcal{L}_{\mathrm{sup}}$ (Cα), then $\mathcal{L}_{\mathrm{CE}}$ head	0.376	0.144
4.2 Freeze after $\mathcal{L}_{\mathrm{sup}}$ (all atoms), then $\mathcal{L}_{\mathrm{CE}}$ head	0.507	0.283
5.1 Freeze after MSE + $\mathcal{L}_{\mathrm{sup}}$ (Cα), then $\mathcal{L}_{\mathrm{CE}}$ head	–	0.170
$\mathcal{L}_{\mathrm{CE}}$ only (benchmark)	0.504	0.307

Table 4: Per-voxel amino-acid classification accuracy under alternative training regimes. $\mathcal{L}_{\mathrm{CE}}$ and $\mathcal{L}_{\mathrm{sup}}$ follow the main-text definitions (cross-entropy and supervised InfoNCE). MSE and cosine are experimental–simulated feature-alignment losses (no residue-type labels). Methods 1.x–3.1 optimize the listed losses jointly; 4.x–5.1 pretrain with $\mathcal{L}_{\mathrm{sup}}$ (optionally plus MSE alignment), freeze the encoder, and fine-tune only a $\mathcal{L}_{\mathrm{CE}}$ classification head. "Cα" restricts $\mathcal{L}_{\mathrm{sup}}$ sampling to Cα centres; "all atoms" uses all atomic centres.

11 Sequence identification from sequence-free models

Sequence identification starts from atomic models built without any supplied sequence (sequence-free / noseq Cα models). In our implementation the sequence-free Cα model used for search is the EModelS (noseq) build. Chains of length at least 30 residues are extracted in both forward and reverse orientations. Residue-type log-probabilities at Cα positions are converted into profile HMMs and searched against UniProtKB with `hmmsearch`; as a single-sequence control, the extracted chain sequences are searched with `phmmer` against the same database. Candidate hits are ranked by E-value and validated by local alignment against the deposited sequence. Figure 15 summarizes this pipeline: model-derived queries, database search, identified sequences, and comparison with reference sequences, including illustrative Smith–Waterman versus Needleman–Wunsch identity bookkeeping.

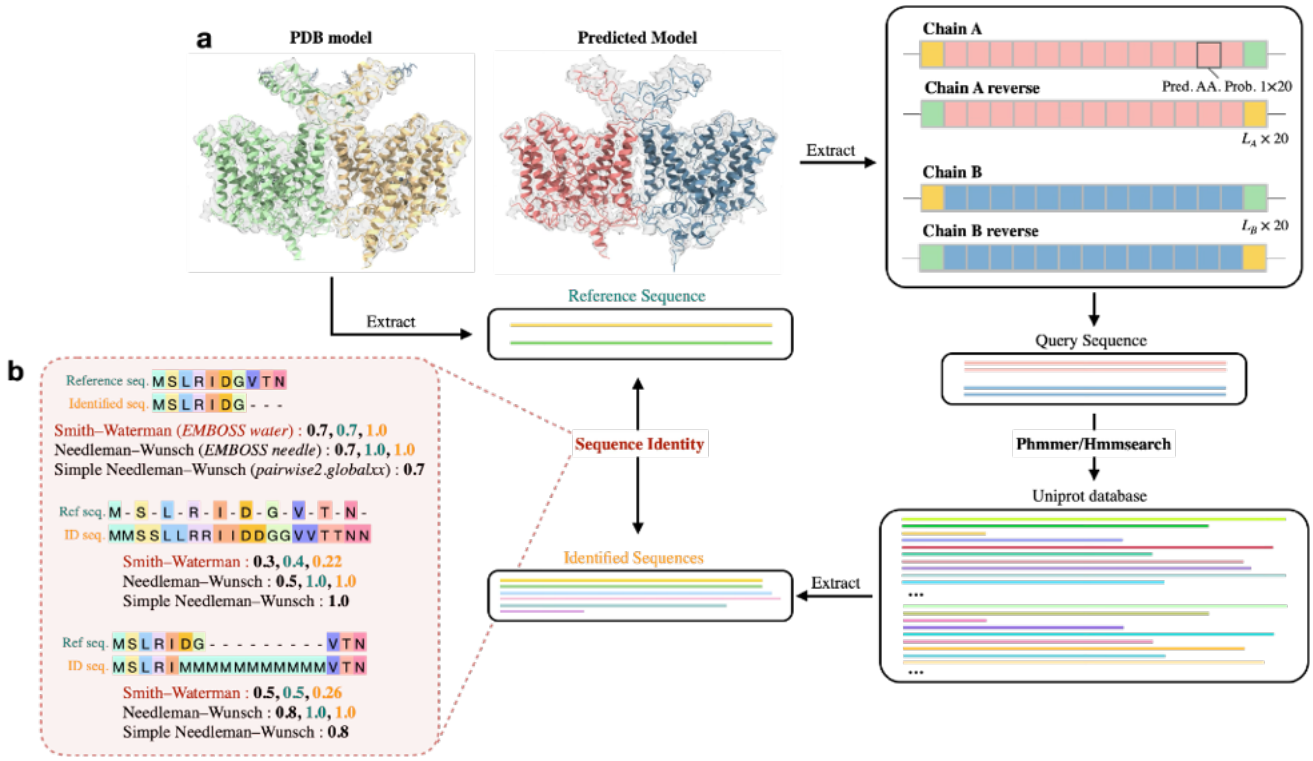


Figure 15: Sequence-search and alignment-based evaluation schematic. **a,** Sequence-free model and reference structure. **b,** Toy alignments comparing local Smith–Waterman (EMBOSS `water`) with global Needleman–Wunsch (EMBOSS `needle`). In each triplet, black is the similarity score (identical aligned residues divided by the full deposited reference length), green is the fraction of the deposited reference sequence covered by the alignment, and yellow is the fraction of the identified (query) sequence covered by the alignment. **c,** Extraction of chain queries from the model. **d,** Database search with `phmmer`/`hmmsearch` yielding identified sequences.